

CHAPTER 8

Sound-Absorbing Materials and Sound Absorbers*

KEITH ATTENBOROUGH

University of Hull
United Kingdom

ISTVÁN L. VÉR

Consultant in Acoustics, Noise, and Vibration Control
Stow, Massachusetts

8.1 INTRODUCTION

One of the most frequent problems faced by noise control engineers is how to design sound absorbers that provide the desirable sound absorption coefficient as a function of frequency in a manner that minimizes the size and cost, does not introduce any environmental hazards, and stands up to hostile environments such as high temperatures, high-speed turbulent flow, or contamination. The designer of sound absorbers must know how to choose the proper sound-absorbing material, the geometry of the absorber, and the protective facing. The theory of sound-absorbing materials and sound absorbers has progressed considerably during the last 10 years. Much of this progress is documented in K. U. Ingard's *Notes on Sound Absorption Technology*¹ and in F. P. Mechel's *Schall Absorber (Sound Absorbers)*² and the separately sold computer program on CD-ROM.³ Ingard's book is a paperback edition of modest length and price and comes with a CD-ROM that allows the reader to make easy numerical predictions from almost all of the equations used in the book. The underlying physical processes are explained with great clarity and the mathematical treatment is kept simple. Derivations and difficult explanation are presented in appendices. The books cover all aspects of sound absorption in great detail. To derive full benefit, it helps if the reader has a reasonably strong mathematical background. However, even those with no such background will find the figures, which give the

*The authors acknowledge the contributions of Dr. Fridolin P. Mechel to the several parts of this chapter retained from the 1992 edition.

difficult mathematical results in graphical form, useful and practical. Mechel's book consists of three hard-covered volumes totaling 2866 pages. The associated computer program is available in MAC and Window versions but is expensive. The classical book of H. Kuttruff, *Room Acoustics*,⁴ deals predominantly with the room acoustics aspects of sound absorption and should be studied by acousticians designing buildings for the performing arts. Beranek's recently updated classic book *Acoustic Measurements*⁵ describes the methods and experimental hardware for measuring all acoustical descriptors of sound-absorbing materials and sound absorbers. The reader with serious interest in sound absorption technology is advised to study all of these texts.

How Sound Is Absorbed

Sound is the organized superposition of particle motion on the random thermal motion of the molecules. The speed of the organized particle motion in air is typically six orders of magnitude smaller than that of the thermal motion. All sound absorbers facilitate the conversion of the energy carried by the organized particle motion into random motion. All forces, other than those that compress and accelerate the fluid, caused by the oscillatory particle flow in the presence of solid material result in loss of acoustical energy. The most important contribution to the conversion is associated with the drag forces caused by friction between the interface of a rigid or flexible wall or the skeleton of the porous or fibrous sound absorber material and the fluid in the thin acoustical boundary layer. For porous and fibrous sound-absorbing materials the acoustical flow speed is low and the flow remains laminar, and the drag force is proportional to the acoustical particle velocity. At high velocities, which occur at the mouth of resonators, the flow separates, turbulence is created, and the friction force becomes proportional to the square of the velocity. Other loss mechanisms, discussed in more detail elsewhere,^{1,2} include the isothermal compression of air at low frequencies, direct conversion of acoustical energy into heat owing to a time lag between compression, and heat flow that takes place in closed cell foams. In the case of plate and foil (very thin plate) absorbers the acoustical energy is converted into heat in the vibrating flexible plate and radiated as sound from the rear of the plate or is transmitted in the form of vibration energy into connected structures.

A sound absorber can absorb only that part of the incident sound energy not reflected at its surface. Consequently, it is important to keep the reflection at the surface as low as possible. Essentially, the part of the incident acoustical energy that enters the absorber should be dissipated before it returns to the surface after traversing the absorber and reflecting from a rigid backing. Otherwise, the absorber gives back acoustical energy to the fluid on the receiver side that is in addition to the initial reflection. This requires a sufficient thickness. The challenge in sound absorber design is to keep the absorber thickness to a minimum.

Sound Absorption Coefficients

The acoustical performance of flat sound absorbers is characterized by their sound absorption coefficient α , defined as the ratio of the sound power, W_{nr} , that is not

reflected (i.e., dissipated in the absorber, transmitted through the absorber into a room to its rear, or conducted, in the form of vibration energy, to a connected structure) and the sound power incident on the face of the absorber, W_{inc} :

$$\alpha \equiv \frac{W_{\text{nr}}}{W_{\text{inc}}} \quad (8.1)$$

For convenience in analyses, the absorption coefficient is defined in terms of sound pressure reflection factor R of the absorber interface, namely

$$\alpha = 1 - |R|^2 \quad (8.2)$$

The vertical lines bracketing R indicate the absolute value. The reflection factor R is usually a function of the angle of sound incidence, the frequency, the material, and the geometry of the absorber. The absorber is characterized by its wall impedance (otherwise known as surface impedance) Z_w , defined as

$$Z_w = \frac{p}{v_n} \quad \text{N} \cdot \text{s/m}^3 \quad (8.3)$$

where p is the sound pressure and v_n is the normal component of the particle velocity, both evaluated at the interface. A substantial part of this chapter concerns the prediction of the wall impedance offered by a large variety of sound absorbers. The sound absorption coefficient as defined in Eqs. (8.1) and (8.2) is further differentiated according to the angular composition of the incident sound field, such as *normal incidence*, *oblique incidence*, and *random incidence*, and whether the absorber is *locally reacting* (sound cannot propagate in it parallel to the interface) or *non-locally reacting* (where sound can propagate in it parallel to the interface). At normal incidence there is no difference between locally and non-locally reacting materials.

Measurement of Normal-Incidence Sound Absorption Coefficient

α_0 . The normal incidence sound absorption coefficient α_0 can be measured in an impedance tube according to the American Society for Testing and Materials (ASTM) standard C384-98 as $\alpha_0 = 4\zeta/(\zeta + 1)^2$, where $\zeta = p_{\text{max}}/p_{\text{min}}$ is the ratio of the maximum and minimum standing-wave sound pressure pattern in the tube upstream of the sample. Normal-incidence sound absorption coefficients, measured in an impedance tube, never exceed unity.

Measurement of Random-Incidence Sound Absorption Coefficient

$\alpha_R(\text{rev})$ in Reverberation Room. The random-incidence sound absorption coefficient $\alpha_R(\text{rev})$, which is also referred to as the *Sabine absorption coefficient*, can be measured directly in a reverberation room according to ASTM C423-02. It is defined as $\alpha_R(\text{rev}) \equiv (55.3V/S)[(1/T_S) - (1/T_0)]$, where V is the volume of the reverberation chamber in m^3 , $S = 6.7 \text{ m}^2$ is the standardized surface area of the test sample, and T_S and T_0 are the reverberation times in seconds (see

Chapter 7) measured with and without the sample, respectively. It is important to distinguish between the two random-incidence sound absorption coefficients $\alpha_R(\text{rev})$ and α_R . The coefficient $\alpha_R(\text{rev})$, measured in the reverberation room, can yield values which exceed unity (measured values up to 1.2, as can be seen in Fig. 8.10). Such values obviously violate the theoretical definition of α given in Eq. (8.36) implying that the panel absorbs more energy than is incident on it. This fact that measured values of $\alpha_R(\text{rev}) > 1$ results from the finite size of the test sample, which means that there is diffraction at the edges of the sample. In design calculations, it is customary to replace all values of $\alpha_R(\text{rev}) > 1$ with unity. The random-incidence sound absorption coefficient α_R is computed from the wall impedance and never yields absorption coefficients that exceed unity. Based on the analyses of $\alpha_R(\text{rev})$ measured according ASTM C423-02 for a large variety of thicknesses and flow resistivity of Owens Corning series 700 fiberglass boards, Godfrey⁶ has proposed an empirical prediction scheme to relate $\alpha_R(\text{rev})$ of a standard size (6.7 m²) sample to α_R predicted theoretically from the computed wall impedance of the samples as

$$\alpha'_R(\text{rev}) = \left(\frac{21.3}{f^{0.5}} + 0.73 \right) \alpha_R$$

where $\alpha'_R(\text{rev})$ is the empirically predicted Sabine absorption coefficient of a standard size sample and f is the frequency in hertz. This formula provides reasonably accurate predictions for low flow resistivities (in the range of $9000 \text{ N} \cdot \text{m/s}^4 = 0.56 \rho_0 c_0 / \text{in.}$ to $17,000 \text{ N} \cdot \text{m/s}^4 = 1.1 \rho_0 c_0 / \text{in.}$) and small layer thicknesses from 1 to 3 in. (2.5–7.5 cm). For flow resistivities greater than $32,000 \text{ N} \cdot \text{m/s}^4 = 2 \rho_0 c_0 / \text{in.}$ and layer thicknesses $\geq 76 \text{ mm} = 3 \text{ in.}$, the analytical method⁷ yields the best prediction.

8.2 SOUND ABSORPTION BY NON-SOUND ABSORBERS

Although inefficiently, all rigid and flexible structures absorb sound. If efficient sound absorbers are present in the room, the contribution of these marginal sound absorbers may be neglected. However, if they represent the only sound-absorbing mechanism (such as is in the case of reverberation rooms without low-frequency absorbers), they, and at high frequencies the air absorption, account for the total sound absorption and set an upper limit for the maximum achievable reverberation time and for the buildup of reverberant sound pressure in these special rooms.

Sound Absorption of Rigid Nonporous Wall

The general assumption that a rigid, nonporous wall, however massive, gives rise to total reflection of the incident sound is not valid. In the immediate vicinity of the interface there are two phenomena that cause small but finite dissipation of sound energy. The first is that the wall-parallel component of the particle velocity of the incident sound results in shear forces in the acoustical boundary

layer. The acoustical boundary layer is identical to that which would develop at the interface between a stationary fluid and a plane, rigid wall oscillating with the frequency and velocity amplitude of the wall-parallel component of the sound-induced particle motion. The second component of the unavoidable dissipation results from the large thermal capacity of the wall, which makes it impossible to fully recover the heat energy in the rarefaction phase that was built up in the compression phase. According to reference 4, for randomly incident sound, the combined effect of these two dissipation processes results in a lower limit of the sound absorption coefficient of

$$\alpha_{\min} = 1.8 \times 10^{-4} (f)^{1/2} \quad (8.4)$$

where f is the frequency in hertz. For frequencies of 1 and 10 kHz, Eq. (8.4) yields values of $\alpha_{\min}$ of 0.006 and 0.018, respectively.

Sound Absorption of Flexible Nonporous Wall

Building partitions such as walls, windows, and doors absorb low-frequency sound. The sound power lost, W^{Loss} , when the incident sound interacts with a nonporous, flexible, homogeneous, and isotropic single partition that has another room or the outdoors on the receiver side is made up of three components:

$$W^{\text{Loss}} = W^{\text{ForcedTrans}} + W^{\text{ResTrans}} + W^{\text{ResDiss}} \quad \text{N} \cdot \text{m/s} \quad (8.5)$$

where $W^{\text{ForcedTrans}}$ is the sound power transmitted by the forced bending waves (i.e., mass law portion radiated from the receiver side by the forced bending waves), W^{ResTrans} is the sound power transmitted by the free resonant bending waves, and W^{ResDiss} is the sound power dissipated in the partition by the free resonant bending waves. The first two terms in Eq. (8.5) can be expressed in the form of sound transmission loss, TL, of the partition and the third term as a function of the space-time mean-square value of the sound-induced resonant vibration velocity $\langle v^2 \rangle$ of the plate, yielding

$$W^{\text{Loss}} = W^{\text{inc}} 10^{-\text{TL}/10} + \langle v^2 \rangle \rho_S \omega \eta S \quad \text{N} \cdot \text{m/s} \quad (8.6)$$

where η is the composite loss factor of the plate accounting for energy loss through dissipation in the plate as well as for that lost through structural coupling to neighboring structures at the plate boundaries and S is the surface area of the plate in square meters. The sound absorption coefficient is defined as

$$\alpha = \frac{W^{\text{Loss}}}{W^{\text{inc}}} \quad (8.7)$$

where W^{inc} is the sound power incident on the source side of the partition. As shown in Chapter 10, for random incidence, where the sound energy has equal

probability of angle of incidence, the incident sound power is

$$W^{\text{inc}} = S \left(\frac{\langle p^2 \rangle}{4\rho_0 c_0} \right) \quad \text{N} \cdot \text{m/s} \quad (8.8)$$

where S is the surface area of the partition (one side) in square meters, $\langle p^2 \rangle$ (N^2/m^4) is the mean-square space-time average sound pressure in the source room, ρ_0 is the density of air in kilograms per cubic meter, and c_0 is the speed of sound of the gas at design temperature in meters per second. Using the results of the analyses in Chapter 10, for the power dissipated in the partition by the free bending waves, one obtains the following approximate formula for the random-incidence sound absorption coefficient α^{rand} of a single, flexible, homogeneous, isotropic, nonporous partition:

$$\alpha^{\text{rand}} \approx 10^{(-\text{TL}_{\text{rand}}/10)} + \frac{2\pi\sqrt{12}\rho_0 c_0^3 \sigma_{\text{rad}}}{\rho_M h^2 c_L \omega^2} \quad (8.9)$$

where TL_{rand} = random-incidence sound transmission loss of panel (measured or predicted according to Chapter 10)

ω = angular frequency, $= 2\pi f$

f = frequency, Hz

σ_{rad} = radiation efficiency of free bending waves (see Chapter 10)

ρ_M = density of plate material, kg/m^3

h = plate thickness, m

c_L = speed sound in plate material, in m/s

The second term on the right side of Eq. (8.9) is strictly valid only if the ratio of the power loss through dissipation and sound radiation by the free bending waves exceeds unity, $(h\rho_M\omega\eta_C/\rho_0c_0\sigma_{\text{rad}}) > 1$, which is almost always the case because σ_{rad} strongly decreases with decreasing frequency below the coincidence frequency of the plate. In Eq. (8.9) the first term represents the loss of energy by the transmission of sound into the receiver space as defined in Chapter 10. The second term represents the combined contribution of the dissipation of the free resonant bending waves in the plate and energy loss in the form of vibration energy transmission to neighboring structures at the boundaries. It is interesting to note that this second term does not depend of the composite loss factor η_C of the plate (the TL of the plate of course depends on η_C in the frequency range at and above the critical frequency where the free bending waves control the sound radiation). The physical reason for this is that in the case of a small loss factor the plate vibrates more vigorously and the combination of the increased vibration and decreased loss factor results in the same dissipation as happens in the case of a larger loss factor and less vigorous vibration response.

While the first term in Eq. (8.9) is valid for any type of partition (including, e.g., double walls and inhomogeneous and non isotropic plates), the second term is valid only for single, homogeneous, isotropic, platelike partitions. In case of

double walls or double windows the first term will exhibit a peak at the double-wall resonance frequency where the TL has a sharp minimum. In the case of double partitions made of two isotropic, homogeneous plates with air space in between, the second term in Eq. (8.9) can be crudely approximated by entering the parameters of the source-side plate into the second term.

8.3 SOUND ABSORBERS USING THIN LAYERS

For the purposes of this chapter, we call those constructions “sound absorbers” which have been designed on purpose to yield high sound absorption in a narrow or wide frequency band. The rest of this chapter deals with the design of such absorbers. We start with those absorber configurations which can be described by a few directly measurable parameters, such as a thin, flow-resistive layer in front of an air space backed by a rigid wall that can be fully characterized by its flow resistance and mass per unit surface area, and will proceed to configurations utilizing thick porous layers and to tuned absorbers.

Thin, Flow-Resistive Layer in Front of Rigid Wall

A sound absorber that consists of a thin, flow-resistive material in front of a rigid wall with an air space in between, such as shown in Fig. 8.1, is the configuration for which it is the easiest to predict acoustical performance. Accordingly, it is logical to treat it first. The flow-resistive layer might be rigid or flexible.

The sketch on the left in Fig. 8.1 depicts a configuration where the air space between the rigid porous layer and the rigid wall is partitioned in a honeycomb pattern to prohibit propagation of sound parallel to the plane of the porous layer. This absorber configuration is called *locally reacting* because the sound field in the absorber depends only on the sound pressure on the interface at the location of the particular honeycomb cell. The sound in the absorber can propagate only perpendicular to the plane of the interface. The sketch on the right in Fig. 8.1 has no partitioning of the air space and the sound can propagate within the absorber parallel to the plane of the interface. The sound pressure at any particular location in the air space depends on the sound pressure on all the locations on the sound-exposed face of the absorber. This configuration is called a *non-locally reacting* sound absorber.

Rigid Porous Layer

A thin, rigid, flow-resistive layer can be fully characterized by a single parameter, namely its flow resistance R_f or its normalized impedance $z' = R_f/(\rho_0 c_0)$, where ρ_0 is the density and c_0 is the speed of sound of the fluid. The normalized impedance of the partitioned air space is $z'' = j \cot(kt)$, where $k = 2\pi/\lambda$ is the wavenumber, λ is the wavelength of the sound, and t is the depth of the air space.

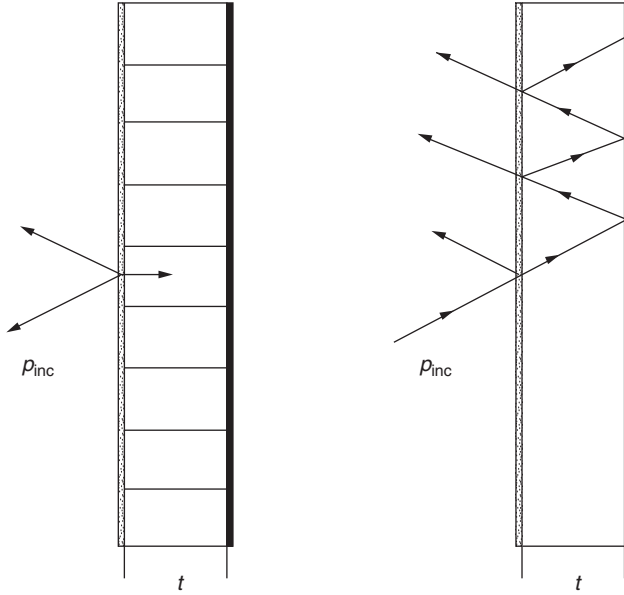


FIGURE 8.1 Thin porous sound-absorbing layer in front of an air space backed by a rigid wall. Left: Partitioned, locally reacting air space. Right: Not partitioned, non–locally reacting air space.

The normalized wall impedance for normal incidence, $\theta = 0$, is then given by

$$z_{w0} = z' + jz'' = \frac{R_f}{\rho c} + j \cot(kt) \quad (8.10)$$

The impedance of the air space is zero when $\cot(kt) = 0$, which is the case at those frequencies where the depth of the air space is an odd number of quarter wavelengths. The maximum value of the normal-incidence sound absorption coefficient is

$$\alpha_{0,\max} = \frac{4z'}{(1 + z')^2} \quad (8.11)$$

indicating that total absorption occurs for $z' = 1$. According to reference 4, the normal-incidence sound absorption coefficient is given by

$$\alpha_0(f) = \left\{ \left[\left(\frac{R_f}{\rho_0 c_0} \right)^{0.5} + \left(\frac{\rho_0 c_0}{R_f} \right)^{0.5} \right]^2 + \frac{\rho_0 c_0}{R_f} \cot^2 \left(\frac{2\pi f t}{c_0} \right) \right\}^{-1} \quad (8.12)$$

Ingard¹ has computed the sound absorption coefficient of a rigid, flow-resistive layer in front of a locally reacting (partitioned) and non–locally reacting

(unpartitioned) air space backed by a rigid wall for both normal and random incidence, as shown in Fig. 8.2.

Figure 8.2 shows that for a locally reacting air space the maximum normal-incidence sound absorption coefficient occurs at those frequencies where the depth of the air space corresponds to an odd multiple of quarter wavelengths [$t = (n + 1)\lambda/4$, where $n = 0, 1, 2, \dots$]. The physical reason for this is that the sound pressure in the partitioned air space just behind the porous layer is zero at these frequencies (the sound that is fully reflected from the hard wall is 180° out of phase with the incident sound). Consequently, the sound pressure gradient across the porous layer, Δp , reaches its maximum value that is numerically equal to the sound pressure on the incident side of the layer, namely $\Delta p = p_{\text{inc}}(1 + R)$, where p_{inc} is the amplitude of the incident sound and R is the reflection factor:

$$R = \frac{R_f - \rho_0 c_0}{R_f + \rho_0 c_0}$$

This pressure gradient produces a particle velocity $v = \Delta p/R_f$ through the porous layer and the sound power dissipated per unit surface area of the absorber,

$$W^{\text{diss}} = v \Delta p = \frac{(p_{\text{inc}})^2(1 + R)^2}{R_f}$$

In the case of $R_f = \rho_0 c_0$, $R = 0$ and $W^{\text{diss}} = (p_{\text{inc}})^2/(\rho_0 c_0) = W^{\text{inc}}$, indicating that all of the incident sound energy is dissipated in the porous layer. For values of $R_f \neq \rho_0 c_0$, the sound absorption coefficient still is a maximum but less than unity. The more R_f differs from $\rho_0 c_0$, the smaller is the sound absorption coefficient.

The maxima of the random-incidence sound absorption coefficient for a partitioned air space occur at the same frequencies as those at normal incidence. However, there is no value of R_f that would yield total absorption.

Another significant feature to be noted from Fig. 8.2 is that for a partitioned, locally reacting air space curves for both the normal and random-incidence sound absorption coefficients versus frequency have notches (zero absorption) at frequencies that correspond to an even multiple of half wavelengths. This occurs because at the corresponding frequencies the sound reflected from the hard wall combines at the rear of the porous layer with the incident sound in such a manner that there is no sound pressure gradient across the porous layer and consequently no sound is absorbed at these frequencies. This does not affect the random-incidence sound absorption coefficient obtained for the not partitioned (non–locally reacting) air space, as shown in the curves marked *c*. The advantage of not having notches in curves of the random-incidence sound absorption coefficient versus frequency must be “paid for” by having substantially lower sound absorption at low frequencies. The most important observation from Fig. 8.2 is that it is not possible to achieve a high sound absorption coefficient with a rigid porous layer in front of an air space backed by a rigid wall unless the thickness

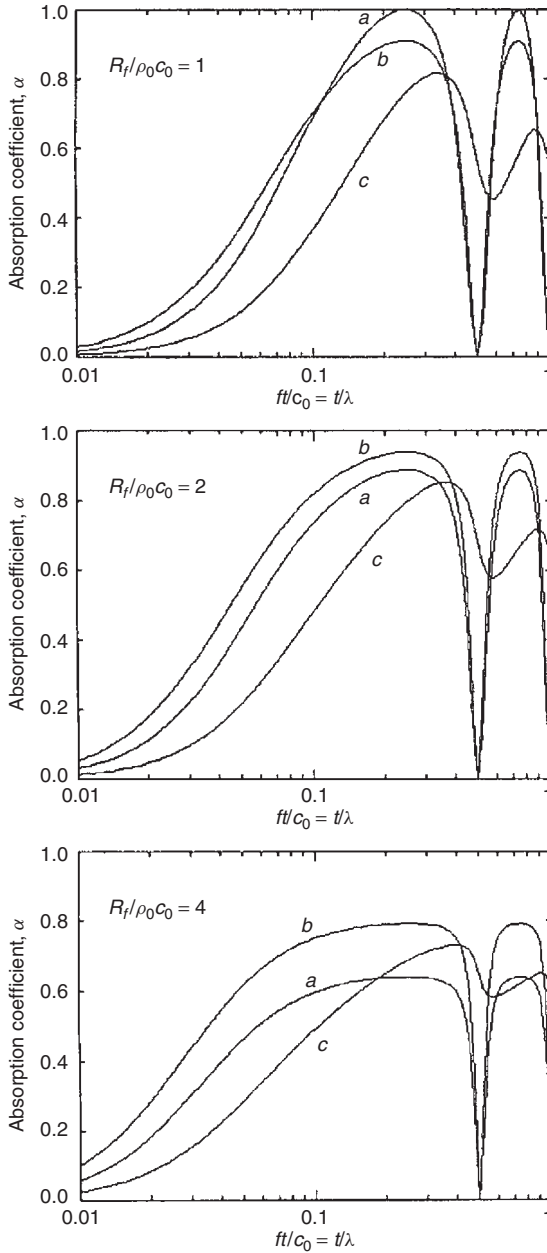


FIGURE 8.2 Sound absorption coefficient of a single, rigid, flow-resistive layer in front of an air space of thickness t backed by a hard wall as a function of the normalized frequency in the form of air space thickness–wavelength ratio $t/\lambda = tf/c_0$ with the normalized flow resistance of the layer, $R_f/\rho_0 c_0$, as the parameter: *a*, normal incidence; *b*, random-incidence, locally reacting air space; *c*, random-incidence, non-locally reacting air space. (After Ref. 1.)

of the air space exceeds one-eighth of the wavelength. For example, an air space thickness of 0.4 m (17 in.) or more is necessary to achieve a high degree of sound absorption at 100 Hz.

As we will discuss below, substantial low-frequency sound absorption can be achieved with an air space thickness of less than one-eighth of the wavelength if the porous layer is not rigid.

Limp Porous Layer

It is little known that sound absorbers using a limp, flow-resistive layer can provide high sound absorption at much lower frequencies than those with a rigid layer. However, caution should be exercised in designing such absorbers for applications where the sound pressure is very high. The limp, flow-resistive layer (e.g., heavy glass fiber cloth or stainless steel wire mesh) should be able to withstand the stresses owing to the high-amplitude sound-induced motion without fatigue. If the porous sound-absorbing layer is limp instead of being rigid, the mass per unit area of the layer and the stiffness per unit area of the air space between the layer and the hard wall result in a resonant system. Near the resonance frequency the limp porous layer will exhibit large-amplitude motion. Approximately, the resonance frequency f_{res} is given by¹

$$f_{\text{res}} \approx \frac{1}{2\pi} \left(\frac{\kappa \rho_0 (c_0)^2}{t m''} \right)^{1/2} \text{ Hz} \quad (8.13)$$

where $\kappa = 1.4$ is the adiabatic compression coefficient, t is the thickness of the air space in meters, and m'' is the mass per unit area of the limp porous layer in kilograms per square meter. The analytical model to predict the sound absorption coefficient of sound absorbers utilizing a limp porous screen is documented in reference 1. The computer program on the CD included in reference 1 allows prediction of the normal- and random-incidence sound absorption coefficient of such absorbers with locally reacting and non-locally reacting air space. The data in Figs. 8.3–8.5 were computed this way and show the sound absorption coefficient as a function of the normalized frequency, $f_n = t/\lambda = tf/c_0$, with the normalized flow resistance of the limp porous layer ($R_f/\rho_0 c_0$) as a parameter for various ratios of the mass of the limp layer and the mass of air in the air space behind, $\text{MR} = m''/t\rho_0$.

Figure 8.3 shows that the first maximum of the normal-incidence sound absorption coefficient occurs close to the mass–spring resonance frequency given by Eq. (8.13). The higher is the mass ratio $\text{MR} = m''/\rho_0 t$, the lower is the frequency of the peak sound absorption. For $\text{MR} = 16$ the frequency of the peak occurs at a frequency that corresponds to a normalized frequency $f_n = 0.03$. This is more than a factor of 8 lower than the peak at $f_n = 0.25$ one would obtain with a rigid porous layer. Also note that the first peak of the normal-incidence sound absorption coefficient is unity (100% absorption) if the mass ratio equals the normalized flow resistance (i.e., when $\text{MR} = R_f/\rho_0 c_0$). To achieve a relatively high

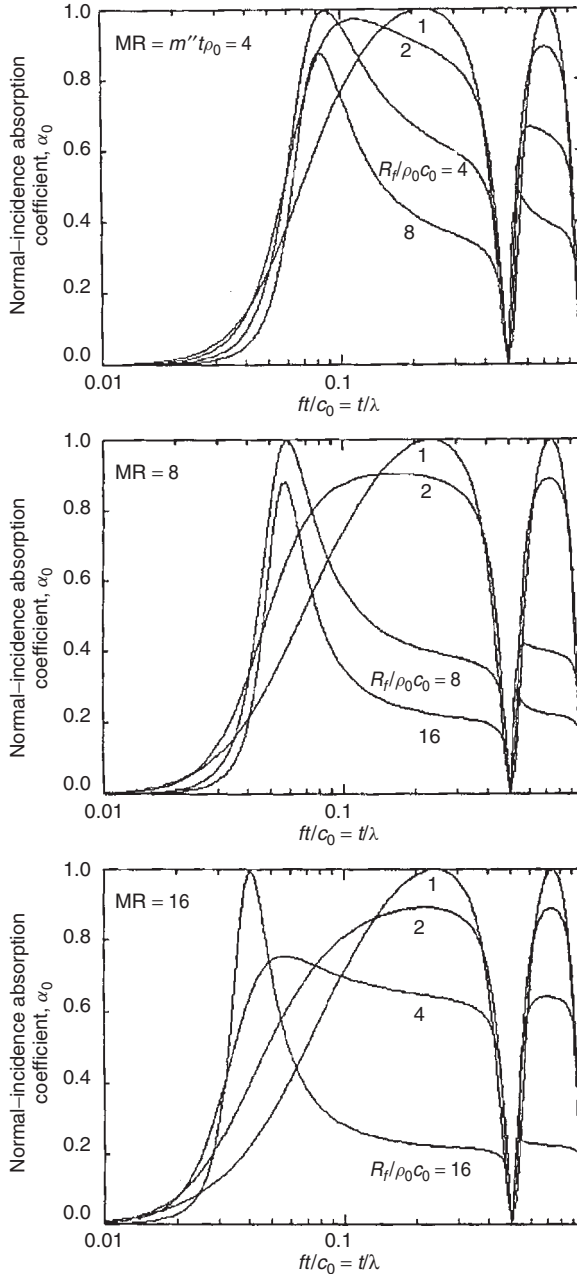


FIGURE 8.3 Normal-incidence absorption coefficient of a single, limp, flow-resistive layer in front of an air space of thickness t backed by a rigid wall as a function of the normalized frequency in the form of the thickness–wavelength ratio $t/\lambda = tf/c_0$ with the normalized flow resistance of the layer, $R_f/\rho_0 c_0$, as the parameter for mass ratios MR . (After Ref. 1.)

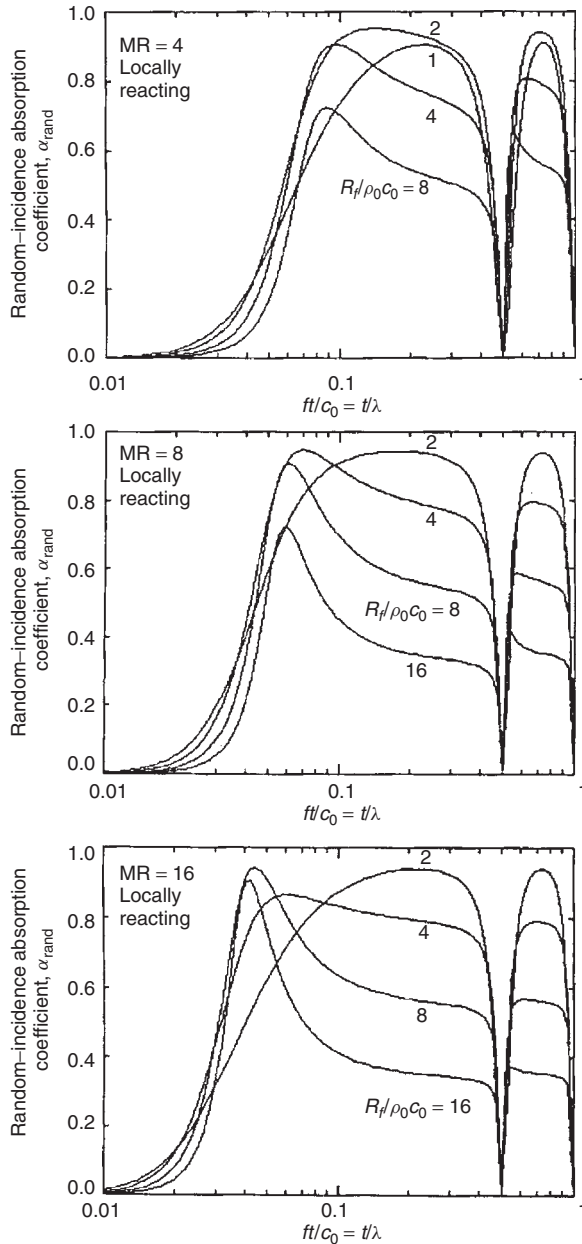


FIGURE 8.4 Random-incidence absorption coefficient of a single, limp, flow-resistive layer in front of a partitioned (locally reacting) air space of thickness t backed by a rigid wall as a function of the normalized frequency in the form of the thickness–wavelength ratio $t/\lambda = tf/c_0$ with the normalized flow resistance of the layer, $R_f/\rho_0 c_0$, as the parameter for mass ratios MR. (After Ref. 1.)

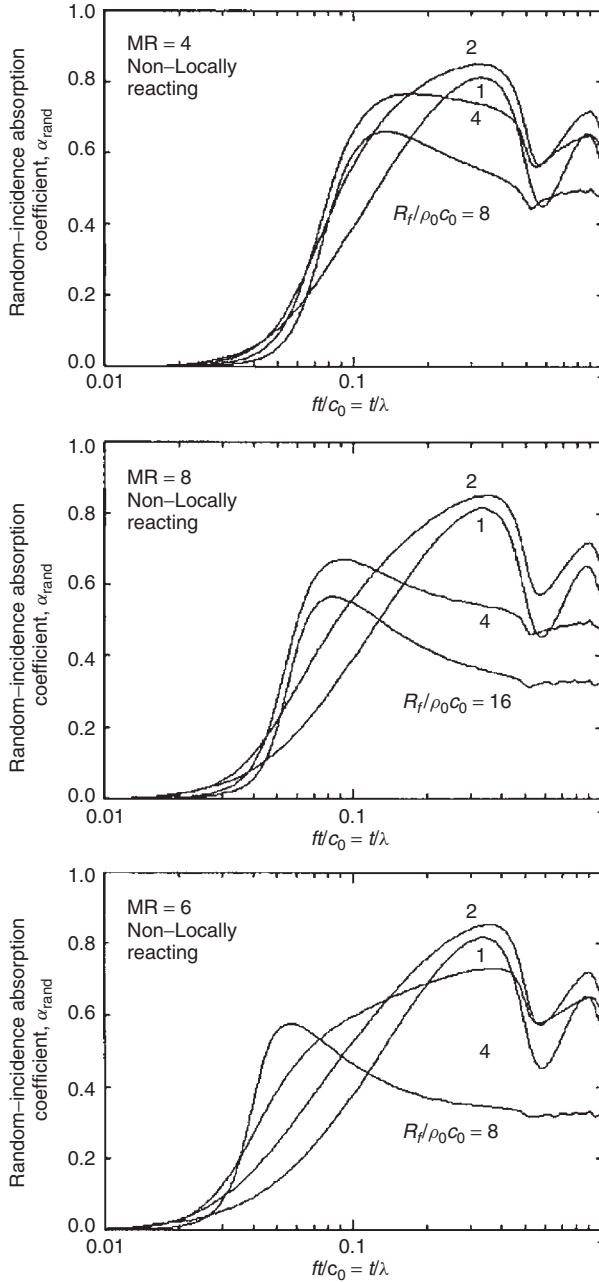


FIGURE 8.5 Random-incidence absorption coefficient of a single, limp, flow-resistive layer in front of a not-partitioned (non-locally reacting) air space of thickness t backed by a rigid wall as a function of the normalized frequency in the form of the thickness-wavelength ratio $t/\lambda = tf/c_0$ with the normalized flow resistance of the layer, $R_f/\rho_0 c_0$, as the parameter for mass ratios MR. (After Ref. 1.)

low-frequency absorption in a wide frequency band, a normalized flow resistance $R_f/\rho_0 c_0$ of between 2 and 3 seems to be a good choice.

Figure 8.4 shows the random-incidence sound absorption coefficient as a function of the normalized frequency computed for a locally reacting (partitioned) air space. Sufficient distance must be provided between the limp porous layer and the honeycomb configuration used to partition the air space to prevent the limp layer from hitting the honeycomb layer at the largest expected displacement amplitude. Curves of the absorption coefficient versus the normalized frequency have similar shape as those in Fig. 8.3 for normal incidence, except that no combination of the design parameters results in total absorption. To obtain substantial low-frequency sound absorption, the mass ratio $MR \approx 2$ seems to be a good choice.

Figure 8.5 shows the random-incidence sound absorption coefficients for a non-locally reacting (unpartitioned) air space. Note the very different behavior from that observed for the partitioned air space in Figs. 8.2 and 8.3. The first, and undesirable, difference is that no combinations of parameters yield random-incidence sound absorption coefficients much above $\alpha_{\text{rand}} = 0.8$. The second, and desirable, attribute of the nonpartitioned air space is that there are no notches in the frequency response where the absorption coefficient would go to zero, as was the case for the partitioned air space.

Multiple Limp Porous Layers

As documented in reference 1, very high sound absorption at low frequencies can be achieved over nearly four octaves by placing a large number (up to 16) of limp, porous layers of low-flow resistance in front of a rigid wall. The computer program supplied with reference 1 allows the prediction of the normal- and random-incidence sound absorption coefficients of such absorbers and the optimization of performance by trial and error varying the flow resistance, mass per unit area, and number and position of the layers.

8.4 POROUS BULK SOUND-ABSORBING MATERIALS AND ABSORBERS

Porous bulk sound-absorbing materials are utilized in almost all areas of noise control engineering. This section deals with the following aspects:

1. description of the key physical attributes and parameters that cause a porous material to absorb sound,
2. description of the acoustical performance of porous sound absorbers used to perform specific noise control functions,
3. compilation of acoustical parameters that allow the quantitative design of sound-absorbing configurations on the bases of material and geometric parameters, and

4. experimental methods for measuring the acoustical parameters of porous sound-absorbing materials and the acoustical performance of porous sound absorbers.

Porous materials used for sound absorption may be fibrous, cellular, or granular. Fibrous materials may be in the form of mats, boards, or preformed elements manufactured of glass, mineral, or organic fibers (natural or man made) and include felts and felted textiles. Applications of fibrous materials in silencers require some form of protective covering. The most common covering consists of perforated metal with a fiberglass cloth behind. Sometimes metal screens are inserted between the perforated metal and fiberglass cloth. Cellular materials include polymer foams of varying degrees of rigidity and porous metals. For special applications increasing use is being made of porous metals (e.g., aluminum foams) as sound absorbers. During manufacture, before solidification, metal foams have closed cells, that is, without interconnections. At this stage such materials are poor sound absorbers. However, as liquid-metal foams solidify, thermal stresses occur. The solidified foams usually have cracked cell walls, which significantly increases sound absorption. In addition, by slightly rolling thin sheets of foam from, say, a thickness of 10 to 9 mm, further mechanical cracking occurs, and the interconnections between adjacent cells widen. This increases sound absorption even more. The result is an absorption coefficient versus frequency that has a maximum between 1 and 5 kHz and a peak value of up to 95%. By placing an air gap between metal foam and a rigid wall, one can shift the frequency curve to lower frequencies. In comparison with other materials, glass wool, for example, which gives high absorption over a wide frequency range, metal foams are not very good sound absorbers. However, the high weight-specific stiffness, good crash-energy absorption ability, and fire resistance of porous metals make them suitable for sound absorption panels in the aircraft and automotive industries.

Granular materials can be regarded as an alternative to fibrous and foam absorbers in many indoor and outdoor applications.⁸ Sound-absorbing granular materials combine good mechanical strength and very low manufacturing costs. Granular materials may be unconsolidated (loose) or consolidated through use of some form of binder on the particles as in wood-chip panels, porous concrete, and pervious road surfaces. As well as man-made granular materials, there are many naturally occurring granular materials, including sands, gravel, soils, and snow. The acoustical properties of such materials are important for outdoor sound propagation.

A common feature of porous sound-absorbing materials is that the pores are interconnected and have typical dimensions below 1 mm, that is, much less than the wavelengths of the sounds of interest in noise control.

Porous materials must be considered to be elastic if they are in the form of flexible sheets. However, in many cases, the stiffness of the solid frame of the material is much greater than that of air so that the material may be treated as if it were rigid over a wide range of frequencies. This means that such materials

can be treated as lossy, homogeneous media. This chapter will be confined to porous materials that may be treated as rigid framed.

How Rigid Porous Materials Absorb Sound

When excited by an incident sound wave, the air molecules in the pores of a porous material are made to oscillate. The proximity of the surrounding solid means that the oscillations result in frictional losses. An important factor that determines the relative contribution of frictional losses is the size (width) of the pores relative to the thickness of the viscous boundary layer. At low frequencies the viscous boundary layer thickness might be comparable with the pore width and the viscous loss is high. At high frequencies the viscous boundary layer thickness may be significantly less than the pore size and the viscous loss is small. At such high frequencies the oscillating flow is “pluglike.” However, the presence of the solid causes changes in the flow direction and expansions and contractions of the flow through irregular pores, resulting in loss of momentum in the direction of wave propagation. This mechanism is relatively important at higher frequencies. In the larger pores and at lower frequencies heat conduction also plays a part in energy loss. During the passage of a sound wave, the air in the pores undergoes periodic compression and decompression and an accompanying change of temperature. If the solid part of the material is relatively heat conducting, then the large surface-to-volume ratio means that during each half-period of oscillation there is heat exchange and the compressions are essentially isothermal. At high frequencies the compression process is adiabatic. In the frequency range between isothermal and adiabatic compression the heat exchange process results in further loss of sound energy. In a fibrous material this loss is especially high if the sound propagates parallel to the plane of the fibers and may account for up to 40% of sound attenuation (energy lost per meter of propagation). The losses from forced mechanical oscillations of the skeleton of a porous material are generally so low that it is reasonable to neglect them.

Physical Characteristics of Porous Materials and Their Measurement

Porosity. The gravimetric measurement of porosity requires the weighing of a known volume of dry material. Since in inexpensive fibrous materials, such as mineral wool, the large droplets (or shot with diameter over 100 μm) may constitute up to 30% of the weight, their contribution to the acoustical porosity should be disregarded. Shot can be separated from fiber by a centrifuge process. The dry weight can be used together with the sample volume to calculate the bulk density ρ_B . Subsequently an assumed solid density is used to calculate the porosity h from

$$h = 1 - \frac{\rho_A}{\rho_B} \quad (8.14)$$

For glass fiber and mineral wool products the density of the fiber material is $\rho_B = 2450 \text{ kg/m}^3$. For silica sand, the density of the mineral grains is 2650 kg/m^3 . A

gravimetric method which may be used with some consolidated granular materials is to saturate the sample with water and deduce the porosity from the relative weights of the saturated and unsaturated samples. Mercury has been used as the pore-filling fluid in some applications (e.g., soils), but for many materials the introduction of liquids affects the pores. Following a proposal of Cremer and Hubert,⁹ Champoux et al.¹⁰ have developed a *dry* method of porosity determination which is based on the measurements of the change in pressure within a sample container subject to a small known change in volume. The lid of the container is a plunger, which is driven by a precise micrometer. The pressure inside the chamber is monitored by a sensitive pressure transducer and an air reservoir connected to the container through a valve serves to isolate the system from fluctuations in atmospheric pressure. The system has been estimated to deliver values of porosity accurate to within 2%. An important feature of the method from the point of view of acoustical properties is that it measures the porosity of connected air-filled pores. However, the gravimetric methods do not differentiate between sealed pores and connected pores. Recently a new method for determining porosity based on a simple measurement of displaced air volume has been proposed.¹¹ This has the advantage that it does not require compensation for temperature, as does the method of Champoux et al.¹⁰

An alternative method which may be used with some consolidated materials is to saturate the sample with water and deduce the porosity from the relative weights of the saturated and unsaturated samples.

An acoustical (ultrasonic) impulse method for measuring porosity using the impulse reflected at the first interface of a slab of air-saturated porous material has been proposed and has been shown to give good results for plastic foams¹²⁻¹⁴ and random bead packing.¹⁵

The porosity of a stack of spherical particles depends on the form of the packing, ranging from 0.26 for the densest packing (face-centered-cubic) to 0.426 for simple-cubic packing.¹⁶ A random packing of spheres has a porosity of 0.356.¹⁷ An approximate value that may be assumed for the porosity of a granular material is 0.4.

Typical ranges of values of porosity are listed in Table 8.1.

Tortuosity. The tortuosity of a porous solid is a measure of the irregularity of the fluid-filled paths through the solid matrix. At very high frequencies, it is responsible for the difference between the speed of sound in air and the speed of sound through a rigid porous material. Tortuosity is related to the formation factor used to describe the electrical conductivity of a porous solid saturated with a conducting fluid. Indeed, tortuosity can be measured using an electrical conduction technique in which the electrical resistivity of such a saturated porous sample is compared to the resistivity of the saturating fluid alone. Thus

$$T = \frac{F}{h} \quad (8.15)$$

TABLE 8.1 Measured and Calculated Values of Porosity and Tortuosity

Material	Porosity, h	Method of Determination	Tortuosity, T	Method of Determination
Lead shot, 3.8 mm particle size	0.385	Measured by weighing	1.6 1.799	Estimated as $\alpha = 1/\phi^{0.5}$ (fits acoustic data) Cell model predictions ^a
Gravel, 10.5 mm grain size	0.45	Measured by weighing	1.55	Deduced by fitting surface admittance data
Gravel, 5–10 mm grain size	0.4	Measured by weighing	1.46	Deduced by fitting surface admittance data
Glass beads, 0.68 mm ^b	0.375	Unspecified	1.742	Measured ^c
<i>Coustone</i> ^b	0.4	Unspecified	1.833	Cell model predictions ^a
Foam YB10 ^b	0.61	Unspecified	1.664	Measured ^c
Porous concrete	0.312	Measured	1.918	Measured ^c
Clay granulate, Laterlite, 1–3 mm grain size	0.52		1.8	Deduced by fitting surface admittance data
Olivine sand	0.444	Measured	1.25	Deduced by fitting surface admittance data (assuming porosity of 0.52)
Aluminum foam	0.93	Measured by weighing	1.626	Cell model predictions ^a
			1.1	Ultrasonic measurements (laser-generated pulses)
			1.07	Deduced by fitting surface admittance data
Polyurethane foam (Recticel Wetter, Belgium)				
Sample w1 ^d	0.98	Measured	1.06	Measured
Sample w2 ^d	0.97	Measured	1.12	Measured

^aFrom Ref. 20.

^bFrom Ref. 26.

^cFrom Ref. 11.

^dFrom Ref. 27.

where F is the formation factor defined by

$$F = \frac{\sigma_s}{\sigma_f} \quad (8.16)$$

where σ_f and σ_s are the electrical conductivities of the fluid and the fluid-saturated sample, respectively. These in turn are defined by

$$\sigma = \frac{GL}{A} \quad (8.17)$$

where L is the length of the sample, A is the area of the end of the sample, and G is the ratio of the resulting current to the voltage applied across the sample.

To measure the formation factor, first a cylindrical sample of the material is saturated with a conducting fluid (brine solution is convenient). Saturation is achieved by drawing the fluid through the sample after forming a vacuum above it. Agitation of the sample is also required if the pore sizes are small. A voltage is applied across the saturated sample placed between two similarly shaped electrodes at a known separation. The conductivity of the fluid is measured at similar voltages within a separate fluid-tight unit. The use of separate current and voltage probes assures a good contact between the end of the sample and the electrodes, eliminates problems associated with voltage drop at the current electrodes, and allows the simultaneous measurement of the electrical resistivities of the fluid and the saturated porous material.

The tortuosity of a random stacking of glass spheres is given by $1/\sqrt{h}$.¹⁸ This has been verified for a range of porosities from 0.33 to 0.38 and is a special case of the relationship

$$T = h^{-n'} \quad (8.18)$$

where n' depends on grain shape and has the value 0.5 for spheres.¹⁹

An alternative derivation for stacked identical spheres²⁰ gives

$$T = 1 + \frac{1-h}{2h} \quad (8.19)$$

The equivalent formula for the tortuosity of a system of identical parallel fibers is

$$T = \frac{1}{h} \quad (8.20)$$

This means that the tortuosity of a typical fibrous material used in noise control is a little larger than 1 since the porosity is close to (but never greater than) 1. In a rigid porous material, tortuosity is one of the properties that contribute to the “structure factor” used in classical descriptions and is related to the “added mass” that is used in the theory of propagation of sound in porous and elastic materials.²¹ The structure factor, introduced in classical texts, is intended to include frequency-dependent thermal effects (complex bulk modulus) and frequency-dependent effects due to fiber motion. This means it can *only* be determined by acoustical means. For characterizing the acoustical properties of a bulk porous material, tortuosity is preferred over the structure factor since it has clear physical meaning and can be deduced by nonacoustical means in many cases.

At the highest frequencies, the speed of sound inside the rigid porous material is equal to the speed of sound in air divided by the square root of tortuosity. Since it is primarily responsible for the acoustical properties of porous materials at high audible or ultrasonic frequencies, tortuosity can be deduced from ultrasonic measurements.²²

Some representative values of tortuosity are listed in Table 8.1.

Flow Resistance and Flow Resistivity. The most important parameter in determining the acoustical behavior of thin porous materials is the airflow resistance R_f . For bulk-, blanket-, or board-type porous materials the flow resistivity (specific flow resistance per unit thickness) $R_1 = R_f/\Delta x$, where Δx is the thickness of the layer, is the key acoustical parameter. The flow resistivity is a measure of the resistance per unit thickness inside the material experienced when a steady flow of air moves through the test sample. Flow resistance R_f represents the ratio of the applied pressure gradient to the induced volume flow rate and has units of pressure divided by velocity. If a material has a high flow resistivity (high flow resistance per unit thickness), it means that it is difficult for air to flow through the surface. In disciplines other than noise control engineering (e.g., geophysics), it is more common to refer to air permeability (k), which is related to the inverse of flow resistivity ($k = \eta/R_1$, where η is the dynamic coefficient of air viscosity). Since flow resistivity is related to the inverse of permeability, high flow resistivity implies low permeability. Typically, low permeability results from very low surface porosity.

Flow resistance is measured as

$$R_f = \frac{\Delta p}{v} = \frac{TS \Delta p}{V} \quad \text{N} \cdot \text{s} \cdot \text{m}^{-3} \text{ or } \text{Pa} \cdot \text{s} \cdot \text{m}^{-1} \text{ (mks rayls)} \quad (8.21)$$

From this the flow resistivity is obtained as

$$R_1 = \frac{R_f}{\Delta x} \quad \text{N} \cdot \text{s} \cdot \text{m}^{-4} \text{ or } \text{Pa} \cdot \text{s} \cdot \text{m}^{-2} \text{ (mks rayls/m)} \quad (8.22)$$

where Δp is the static pressure differential across a homogeneous layer of thickness Δx , v is the velocity of the steady flow through the material, V is the volume of air passing through the test sample during the time period T , and S is face area (one side) of the sample.

Since R_f generally depends on the velocity v , it is customary to measure it at a number of different flow rates and extrapolate measured R_f versus v to $R_f(v = 0.05 \text{ cm/s})$ because below this particle velocity the flow resistance of most fibrous materials does not depend any more on the velocity. The fact that, in general, flow resistivity depends on particle velocity becomes important when considering the behavior of porous absorbers at high sound levels.

The measurement of the flow resistance and flow resistivity of porous building materials has been standardized on a compressed-air apparatus.²³ A similar apparatus may be used to determine the flow resistivity of soils or granular materials (Fig. 8.6). In this measurement the pressure gradient across the sample in a fixed sample holder is monitored together with various (low) flow rates. Compressed air is passed through a series of regulating valves and a very narrow opening into chamber E. This creates an area of low pressure immediately in front of the three tubes connected to the rest of the system. Air is drawn from the environment through the sample as a result of the pressure differential. The rate of airflow through the system is controlled by three flowmeters, giving a total measurement

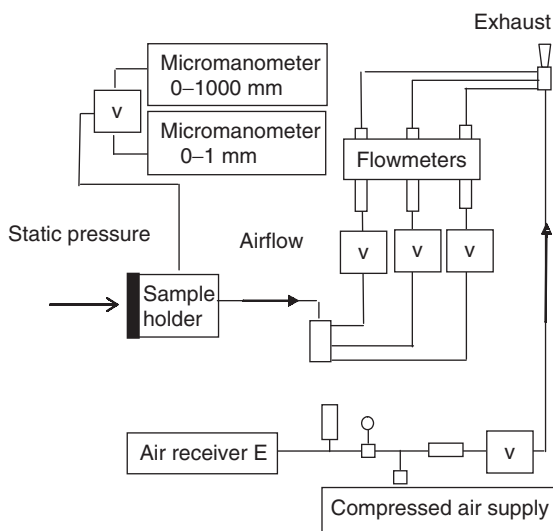


FIGURE 8.6 Concept sketch of a compressed-air apparatus for laboratory measurement of flow resistance.

range between 8.7 and 0.1 L/min. Typically the flow rate must be kept below 3 L/min to avoid structural damage to the sample.

A comparative method²⁴ makes use of a calibrated known resistance (a laminar-flow element) placed in series with the test sample. Variable-capacitance pressure transducers are used to measure pressure differences across both the test sample and the calibrated resistance. For steady, nonpulsating flow, the ratio of flow resistances equals the ratio of measured pressure differences. The airflow may also be controlled electronically. A unique method of measuring flow resistance by Ingard¹ does not require any flow-moving device, flow-rate meter, or static pressure differential sensor, just a stop watch. In this test setup the airflow (after the piston reaches its terminal velocity) is forced through the sample, located at the open bottom end of the tube, at a constant rate driven by the static pressure differential $\Delta p = Mg/S$, determined by the weight, Mg , of a tightly sliding piston in the sample holder tube and by the cross-sectional area S of the piston. The factor $g = 9.81 \text{ m/s}^2$ is the acceleration due to gravity. The flow resistance of the sample is then inversely proportional to the time T_L required for the piston to travel a specific distance L . The flow resistance of the sample is determined as $R_f = C[Mg \cos(\Phi)/LA^2]ST_L$, where A is the free surface area of the sample and Φ is the angle of tube axis with the vertical. For vertical tube orientation $\Phi = 0$. The small leakage flow between the piston and the tube wall, which is of consequence only if the sample has a very high flow resistance, is accounted for by the correction factor

$$C = \frac{1 + T_{L0}/T_L}{1 - T_L/T_{\text{Closed}}}$$

where T_{L0} and T_{Closed} are respectively the piston travel times with the sample removed and when the end of the tube is air tightly closed. The variation of the flow rate through the sample (needed to extrapolate the flow resistance to 5×10^{-4} m/s) can be accomplished by either changing the tube axis angle Φ or increasing the mass of the piston.

Empirical Prediction of Flow Resistivity. According to private communications with Richard Godfrey⁶ of Owens Corning Corporation, the flow resistivity of fibrous sound-absorbing materials can be approximately predicted by a slightly modified version of Eq. 10.4 of the 1971 edition of this book. The modified equation is

$$R_1 \approx \frac{3450}{d^2} \left[\left(\frac{\text{SpGrGlass}}{\text{SpGrFiber}} \right) \rho_{\text{bulk}} \right]^{1.53} \text{ N} \cdot \text{m/s}^4 \quad (8.23)$$

where d is the average fiber diameter in micrometers (10^{-6} m), SpGrGlass and SpGrFiber are the specific gravities of glass and that of the actual fiber material, and ρ_{bulk} is the bulk density (with shot contribution deducted) of the fibrous material in kilograms per cubic meter. Note that $1 \mu\text{m} = 4 \times 10^{-5}$ in. and that $\rho_{\text{bulk}} \approx \rho_M h$, where ρ_M is the density of the fiber material and h is the porosity. It has been found that flow resistivity values measured according to the American Society of Mechanical Engineers (ASME) Standard E 522 for the entire glass fiber-based product line of different bulk densities and fiber diameters are successfully predictable by Eq. (8.23) with a correlation coefficient of 0.93. Equation (8.23) also can be used for predicting the flow resistivity of polymer-based fibrous sound-absorbing materials. For the same bulk density, the polymer fibers have approximately 2.5 times as much surface area and flow resistivity as glass fibers.

Theoretical Prediction of Flow Resistivity. For almost all granular or fibrous sound-absorbing materials, the flow resistivity cannot be predicted analytically on the basis of the geometry of the skeleton. It must be measured. For a few idealized absorber materials, such as materials made of identical spheres or parallel identical fibers, it is possible to make such a prediction. Since the prediction formulas identify important parameters that might be used to scale flow resistivity data measured for a specific material composition to other compositions in cases where direct measurement of the flow resistivity is not feasible, it is instructive to consider at least one idealized case where theoretical prediction is possible.

For stacked identical spheres with radius r it can be shown that²⁰

$$R_1 = \frac{\eta}{k} = \frac{9\eta(1-h)}{2r^2h^2} \frac{5(1-\Theta)}{5-9\Theta^{1/3}+5\Theta-\Theta^2} \quad (8.24)$$

where h is the porosity and

$$\Theta = \frac{3}{\sqrt{2\pi}}(1-h) \cong 0.675(1-h) \quad (8.25)$$

The equivalent formula for the flow resistivity of a system of identical parallel fibers of radius r is²⁵

$$R_1 = \frac{16\eta h(1-h)}{r^2 h^2 [(1-8\Gamma) + 8\Gamma(1-h) - (1-h)^2 - 2 \ln(1-h)]} \quad (8.26)$$

where $\Gamma \approx 0.577$. This formula overestimates the flow resistivity of fibrous materials as a consequence of shot content, random fiber orientation, and distribution of fiber diameters in a real material.

Analytical Characterization of Porous Granular or Fibrous Sound-Absorbing Materials. For certain idealized microstructures such as parallel, identical cylindrical or slitlike pores, it is possible to make straightforward analytical predictions of the acoustical properties.^{1,19,21} Although analytical characterization of porous granular or fibrous sound-absorbing materials with arbitrary microstructures is possible, it is a relatively complex procedure.²⁷⁻³¹ Moreover, at present, many of the required parameters are not routinely available to practicing noise control engineers. Consequently, the analysis is not reviewed in this chapter. For a comprehensive introduction to recent analytical treatments, the reader should consult reference 21. Semiempirical formulas for the acoustical properties of fibrous materials have been derived based on such a sophisticated analytical treatment.³² It should be noted, also, that empirical formulas have been derived that predict the acoustical properties of granular media from knowledge only of porosity, grain density, and mean grain size.³³

Acoustical Properties of Porous Materials. The acoustical properties of porous sound-absorbing materials are characterized by their complex propagation constant k_a and complex characteristic impedance Z_{Ca} , which are defined by

$$p(x, t) = \hat{p} e^{-jk_a x} e^{j\omega t} \quad \text{N/m}^2 \quad (8.27)$$

$$\langle p(x, t) \rangle_y = Z_{Ca} \langle v_x(x, t) \rangle_y \quad \text{N/m}^2 \quad (8.28)$$

where $\hat{p}$ represents the amplitude at $x = 0$ and $\langle \dots \rangle$ represents an average perpendicular to the propagation direction (x) over an area that is small compared with the wavelength but large compared with the size of pores. The propagation constant

$$k_a = \beta - j\alpha$$

includes the attenuation constant α , which may be obtained by using a probe tube microphone to measure the decrease of the sound pressure level (in nepers per meter) of a plane sound wave propagating in a very thick layer of material. The phase constant β is obtained by measuring the change of phase with distance. It is equal to the angular frequency divided by the frequency-dependent sound speed within the material. The complex characteristic impedance $Z_{Ca} = R_{Ca} - jX_{Ca}$ is obtained by measuring the surface impedance of a thick layer (sufficiently thick

that reflection from the end is not detectable) of the absorber material placed in an impedance tube.

An alternative to use of a thick sample with a practically “infinite length” is to use a sample of known finite length and to measure the transfer function between two microphones located in an impedance tube with the loudspeaker source at one end and the sample at the other (closed) end.³⁴ A broadband input signal such as white noise or a sine sweep can be used. By means of a frequency analysis, output signals from the microphones are used to calculate the transfer function, which is converted to the surface impedance of a sample. The characteristic impedance and propagation constant of a sample can be obtained as long as two distinct sets of surface impedance are measured. This can be realized by using either two samples of different thicknesses³⁵ or a single sample backed with two different lengths of air cavities.³⁶ For the former, the two-thickness method, it is convenient if the length of the second sample is double that of the first. In the two-cavity method, the difference in the lengths of the air cavities needs to be tuned to fit the frequency range of interest together with the tube diameter and the microphone spacing.

Empirical Predictions from Regression Analyses of Measured Data

There are two families of parameters that determine the sound absorption coefficient. The characterization would be simplest if both k and Z_C could be expressed in terms of a single parameter. Figure 8.7a shows the measured³⁷ normal-incidence sound absorption coefficient α_0 of different Rockwool materials of practically infinite thickness (between 0.5 and 1 m, depending on the bulk density of the absorber material) as a function of frequency with bulk density of the material as parameter. Figure 8.7a clearly indicates that the bulk density is not the parameter that would collapse the measured data points into a single curve.

In 1970, Delany and Bazley³⁸ used many measurements on fibrous materials to deduce semiempirical formulas based on the dimensioned parameter (f/R_1) . Equations (8.29a) and (8.29b) are valid only if the frequency f is entered in hertz (s^{-1}) and the flow resistivity R_1 in mks rayls per meter ($\text{N} \cdot \text{s} \cdot \text{m}^{-4}$). Although they have been used widely and successfully, Delany and Bazley’s formulas give rise to unphysical results at low frequencies. In particular, the real part of the surface impedance of a rigid-backed layer goes negative. Miki’s model³⁹ represents a modification of these formulas based on Delany–Bazley’s data and an electrical analogy for the acoustical properties of porous materials. Miki’s formulas are as follows:

$$k_{an} = \frac{k}{k_0} = 1 + 0.109 \left(\frac{\rho_0 f}{R_1} \right)^{-0.618} - j0.160 \left(\frac{\rho_0 f}{R_1} \right)^{-0.618} \quad (8.29a)$$

$$Z_{Cn} = \frac{Z_C}{Z_0} = 1 + 0.070 \left(\frac{\rho_0 f}{R_1} \right)^{-0.632} - j0.107 \left(\frac{\rho_0 f}{R_1} \right)^{-0.632} \quad (8.29b)$$

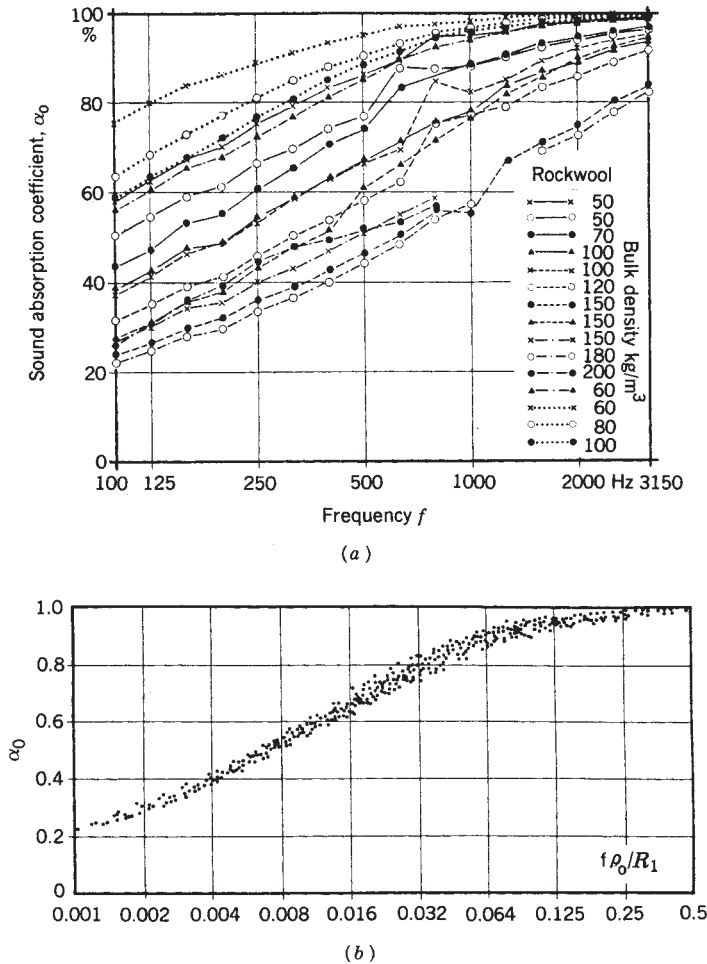


FIGURE 8.7 Normal-incidence sound absorption coefficients of different rock wool materials of practically infinite thickness (0.5–1 m) measured in an impedance tube plotted as a function of (a) frequency with bulk density ρ_A as parameter and (b) nondimensional frequency parameter $\rho_0 f / R_1 = \rho_0 c_0 / (R_1 \lambda)$.

Figure 8.7b shows the same data points as Fig. 8.7a but now as a function of the *dimensionless* variable $E = \rho_0 f / R_1$ on the horizontal scale, where ρ_0 is the density of air, f is the frequency, and R_1 is the flow resistivity of the bulk material at the density at which α_0 was measured. Clearly $E = \rho_0 f / R_1 = \rho_0 c_0 / \lambda R_1$ is the single parameter that collapses all measured data. According to Fig. 8.7b, even for a layer of practically infinite thickness, very high sound absorption is obtainable only in the frequency range where $(\lambda/4)R_1 < \rho_0 c_0$. To fulfill this requirement at low frequencies, where λ is large, requires the use of a sound-absorbing material of low flow resistivity. The normalized, dimensionless frequency variable

$E = \rho_0 f / R_1$ is useful in describing not only the sound-absorbing capability of the semi-infinite layer of fibrous material but also the propagation constant and characteristic impedance of the bulk material.

It is useful to present the propagation constant and characteristic impedance in a dimensionless manner as

$$k_{an} = \frac{k}{k_0} = \frac{a - jb}{k_0} = a_n - jb_n \quad (8.30a)$$

$$Z_{Cn} = \frac{Z_C}{Z_0} = R - jX \quad (8.30b)$$

where $k_0 = \omega / c_0$ is the wavenumber in air and $Z_0 = \rho_0 c_0$ is the characteristic impedance of the gas filling the voids between the fibers for plane waves.

Figures 8.8 and 8.9 show plots of the real and imaginary parts of the normalized propagation constant k_{an} and that of the normalized characteristic impedance Z_{Cn} for a large variety of mineral wool sound-absorbing materials plotted as a function of the normalized frequency parameter, indicating that indeed the normalized frequency parameter $E = \rho_0 f / R_1$ is a universal descriptor of fibrous porous sound-absorbing materials. The data presented in Figs. 8.8 and 8.9 and similar curves for glass fiber materials were obtained by careful measurements of the acoustical and material characteristics (a , b , R , X , and R_1) of over 70 different types of materials.³⁸

The solid lines in Figs. 8.8 and 8.9 result from the regression analyses of the data and have the form

$$k_{an} = \frac{k}{k_0} = (1 + a'' E^{-\alpha''}) - ja' E^{-\alpha'} \quad (8.31a)$$

$$Z_{Cn} = \frac{Z_C}{Z_0} = (1 + b' E^{-\beta'}) - jb'' E^{-\beta''} \quad (8.31b)$$

The regression parameters a' , a'' , b' , b'' , α' , α'' , β' , and β'' in Eqs. (8.31) are compiled in Table 8.2. There are different regression parameters for the normalized frequency regions below and above $E = 0.025$. It was found that measured fibrous materials could be divided into two categories: (1) mineral wool and basalt wool and (2) glass fiber.

Polyester Fiber Materials

Polyester fiber materials are used increasingly to replace glass and mineral fiber materials in situations where there is concern to keep the air completely free of fibers suspected to have an adverse influence on health. An example material consists of a mix of two kinds of fibers: (1) polyethyleneterephthalate and (2) a core of polyethyleneterephthalate and a lining of copolyester. The raw mix is treated at 150°C to melt the external lining of the “bicomponent” fibers and hence form a skeleton of thermally bound fibers. The fiber diameters are between

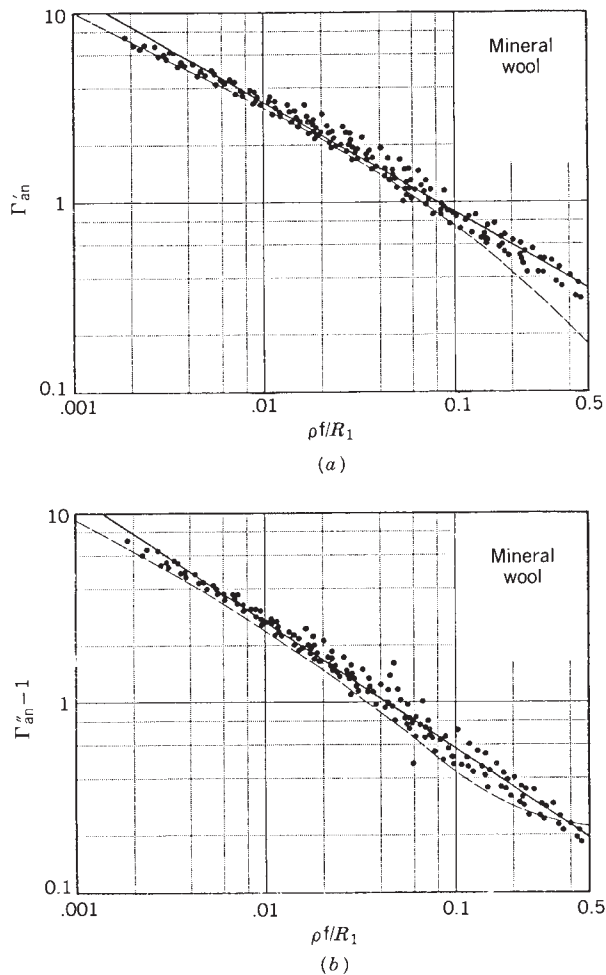


FIGURE 8.8 Measured normalized propagation constant $k_{an} = k/k_0 = a_n + ib_n$ for mineral wool as a function of normalized frequency parameter $E = \rho_0 f/R_1$: (—) regression line: Eq. (8.31a) and Table 8.2; (- - -) prediction for cylindrical pores.^{19,21,27}

TABLE 8.2 Regression Coefficients for Predicting Propagation Constant and Characteristic Impedance of Fibrous Sound-Absorbing Materials

Material	E Region	b'	β'	b''	β''	a''	α''	a'	α'
Mineral and basalt wool	$E \leq 0.025$	0.081	0.699	0.191	0.556	0.136	0.641	0.322	0.502
	$E > 0.025$	0.0563	0.725	0.127	0.655	0.103	0.716	0.179	0.663
Glass fiber	$E \leq 0.025$	0.0668	0.707	0.196	0.549	0.135	0.646	0.396	0.458
	$E > 0.025$	0.0235	0.887	0.0875	0.770	0.102	0.705	0.179	0.674

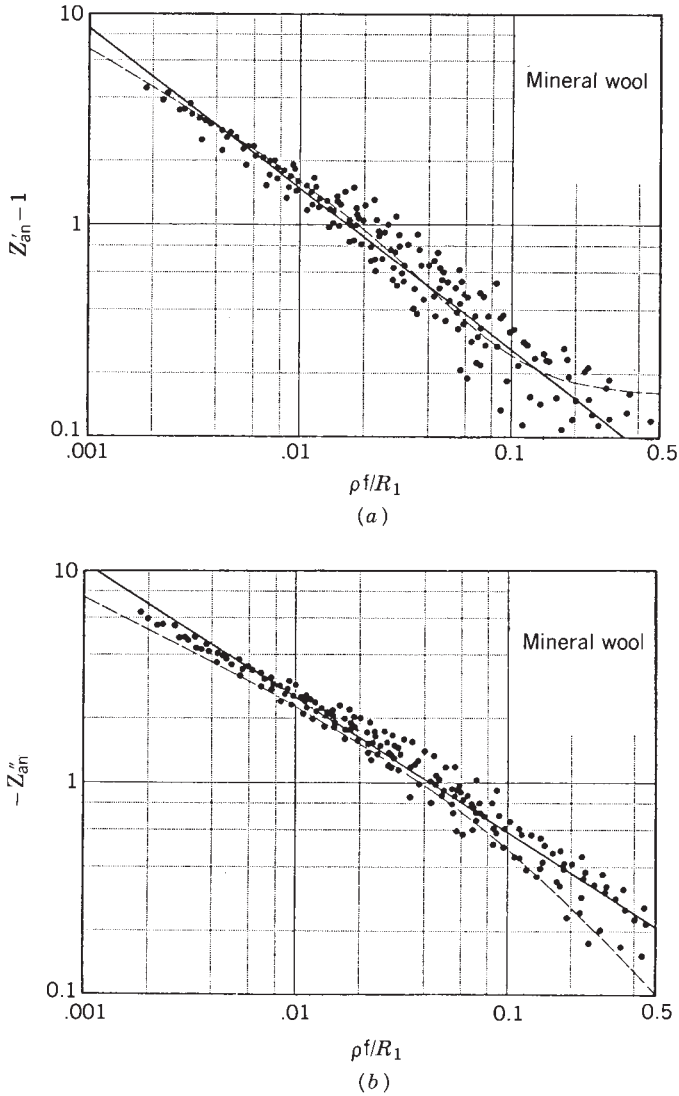


FIGURE 8.9 Measured normalized characteristic impedance $Z_{Cn} = Z_C/Z_0 = R + iX$ for mineral wool as a function of normalized frequency parameter $E = \rho_0 f / R_1$: (—) regression line; Eq. (8.29b); (---) prediction for cylindrical pores.

17.9 and 47.8 μm (mean 33 μm) and have a mean length of 55 mm. Based on measurements of 38 samples, the flow resistivity has been found to obey the relationship⁴⁰

$$R_1 D^2 = 26 \rho_A^{1.404} \quad (8.32)$$

where D is the mean diameter in micrometers and ρ_A the bulk density in kg/m^3 .

TABLE 8.3 Regression Coefficients for Predicting Propagation Constant and Characteristic Impedance of Polyester Fiber Materials

Material	b'	β'	b''	β''	a'	α'	a''	α''
Polyester fiber	0.159	0.571	0.121	0.530	0.078	0.623	0.074	0.660

Regression coefficients for predicting acoustical properties of polyester fiber materials are given in Table 8.3.³⁸

Plastic Foams

Since concerns about fire hazard and release of toxic combustion products can be overcome largely by suitable treatments, plastic foams are used increasingly for noise control applications. A large variety of plastic foams are available with several different types of physical structure. Polyurethane foams, based on polyester or polyether polyols, are used most commonly. These foams can be fully or partially reticulated, that is, with all membranes or with varying proportions of membranes between cells removed. Formulas (8.31) with similar regression coefficients to those for fibrous materials are applicable to plastic foams.^{41,42} However, unlike with fibrous materials, it has not been found necessary to distinguish between E-regions. The relevant regression coefficients are listed in Table 8.4.

The first row of Table 8.4 also shows the comparable values obtained from the empirical formulas of Delany and Bazley (ref. 38). Although these formulas

TABLE 8.4 Regression Coefficients for Predicting Propagation Constant and Characteristic Impedance of Plastic Foams

Material	b'	β'	b''	β''	a'	α'	a''	α''
Mineral and glass fiber	0.0571	0.754	0.087	0.732	0.0978	0.700	0.189	0.595
Fully reticulated polyurethane foam, $60 \leq R_1 \leq 6229$ (Cummings/Beadle)	0.0953	0.491	0.0986	0.665	0.174	0.372	0.167	0.636
Mixed plastic foams, ⁴² $2900 \leq R_1 \leq 24300$	0.209	0.548	0.105	0.607	0.188	0.554	0.163	0.592
Fully reticulated polyurethane foam, $380 \leq R_1 \leq 3200^a$	0.114	0.369	0.0985	0.758	0.136	0.491	0.168	0.795
Partly reticulated polyurethane foam, $R_1 = 10,100^b$	0.279	0.385	0.0881	0.799	0.267	0.461	0.158	0.700

^aFrom ref. 43.

^bFrom ref. 44.

have been superseded by those due to Mechel (ref. 37) and Miki (ref. 39), they serve to indicate that the coefficient values for plastic foams are significantly different and that the Delany and Bazley relationships would not generally give satisfactory predictions for the bulk properties of plastic foams.

Effects of Temperature

In sound absorbers or silencers designed to operate at high temperatures, it is important to know the acoustical characteristics of the fibrous porous sound-absorbing materials at design temperature. Fortunately, it is not necessary to measure the propagation constant k and characteristic impedance Z_C at design temperature T because the values of these acoustical characteristics measured at room temperature, T_0 , can be scaled. The dimensionless frequency variable $E = \rho_0 f / R_1$ must be evaluated at design temperature. Consider that the influence of temperature on ρ and η is

$$\rho(T) = \rho(T_0) \frac{T_0}{T} \quad (8.33)$$

$$\eta(T) = \eta(T_0) \left(\frac{T}{T_0} \right)^{0.5} \quad (8.34)$$

where T and T_0 are the design temperature and room temperature, respectively, measured on an absolute scale (i.e., Kelvin and Rankine; $K = ^\circ C + 273$). Considering that $\rho = \rho_0(T/T_0)^{-1}$ and $c = c_0(T/T_0)^{1/2}$, the acoustical characteristics at design temperature T (in Kelvin) are determined as follows:

$$b(T) = b[E(T)] \frac{\omega}{c_0} \left(\frac{T}{T_0} \right)^{-1/2} \quad (8.35a)$$

$$Z_C(T) = Z_{Cn}[E(T)](\rho c_0) \left(\frac{T}{T_0} \right)^{-1/2} \quad (8.35b)$$

where $b[E(T)]$ and $Z_{Cn}[E(T)]$ are the normalized attenuation constant and normalized characteristic impedance according to Eqs. (8.30) computed for $E = E(T)$ determined from Eqs. (8.34) and (8.35). When fibrous sound-absorbing materials are first used in high-temperature applications, the material undergoes changes. For mats and boards the binder burns off. This decreases the bulk density to a negligible amount. The burning off of the binder does not appreciably change the flow resistivity. After the first high-temperature exposure the glass or mineral fibers thicken. Their virgin diameter d_V increases to the burned-off diameter $d_{BOF} = d_V + 0.5$. This increase is approximately $0.5 \mu\text{m}$. This increase in fiber diameter and corresponding decrease in flow resistivity can be accounted for by correcting flow resistivity measured at room temperature by multiplying it by the factor $(1 + 0.5/d_V)^{-2}$ when calculating $E_{BOF}(T_0) = f\rho_0/R_1(1 + 0.5/d_V)^{-2}$.

Example 8.1. Compute the real and imaginary parts of the normalized propagation constant and characteristic impedance of a very (practically infinite) thick layer of glass fiber at a frequency of 100 Hz at 20°C and for 500°C and determine α_N using the regression parameter values in Table 8.2. The flow resistivity of the fibrous material at 20°C is $R_1(T_0) = 16,000 \text{ N} \cdot \text{s/m}^4$ and $\rho_0 = 1.2 \text{ kg/m}^3$.

Solution

$$\begin{aligned}
 T_0 &= 273 + 20 = 293 \text{ K} & T &= 273 + 500 = 773 \text{ K} \\
 E(T_0) &= \frac{\rho_0 f}{R_1} = 1.2 \times \frac{100}{16,000} = 0.0075, \quad \text{which is } < 0.025 \\
 E(T) &= E(T_0) \left(\frac{T}{T_0} \right)^{-1.65} = 0.0075 \left(\frac{773}{293} \right)^{-1.65} = 1.5 \times 10^{-3}, \\
 &\text{which is } < 0.025
 \end{aligned}$$

The regression parameters from Table 8.2 are

$$\begin{aligned}
 a' &= 0.396 & a'' &= 0.135 & \alpha' &= 0.458 & \alpha'' &= 0.646 \\
 b' &= 0.0668 & b'' &= 0.196 & \beta' &= 0.707 & \beta'' &= 0.549
 \end{aligned}$$

According to Eqs. (8.35), we obtain the following:

Parameter	At 20°C	At 500°C
$b = a' E^{-\alpha'}$	3.72	7.78
$a = 1 + a'' E^{-\alpha''}$ a	4.18	10.0
$R = 1 + b' E^{-\beta'}$ R	3.12	7.62
$X = b'' E^{-\beta''}$	2.88	6.9

The normal-incidence sound absorption coefficient according to Eq. (8.36) is

$$\alpha_N = 4 \frac{Z'_{an}}{Z'^2_{an} + 2Z'_{an} + 1 + Z''^2_{an}}$$

yielding

$$\alpha_N = \begin{cases} 0.49 & \text{for } 20^\circ\text{C} \\ 0.25 & \text{for } 500^\circ\text{C} \end{cases}$$

8.5 SOUND ABSORPTION BY LARGE FLAT ABSORBERS

When considering sound absorption, the portion of a sound wave incident on the absorber that is absorbed is a quantity of interest. This is easiest to define when

the surface of the absorber is flat and sufficiently large that sound waves scattered at the edges of the absorber can be neglected. For a plane incident sound wave, it is possible to assign a sound energy absorption coefficient α for each point on the absorber surface given by

$$\alpha = \frac{\text{absorbed energy}}{\text{incident energy}} = 1 - |R|^2 \quad (8.36)$$

where R is the reflection factor, which is defined as the ratio of the reflected and incident sound pressures at the interface. A high sound absorption coefficient ($\alpha \rightarrow 1$) requires that $|R| \rightarrow 0$. Note that $|R| = 0.1$ corresponds to $\alpha = 0.99$. In this chapter we shall deal only with infinitely large, flat, homogeneous absorbers. Edge effects manifest themselves in increased sound absorption with increasing perimeter–surface area ratio of the absorber.⁴⁵ Edge effects are involved also when numerical values greater than 1 are obtained for random-incidence absorption coefficients measured in reverberation rooms.

Plane Sound Waves at Normal Incidence

For plane waves of sound incident perpendicularly to an absorber it is sufficient to know the complex normal specific *surface impedance* $Z_1 = Z'_1 - jZ''_1$ of the absorber, which is the ratio of sound pressure to the *normal* component of the particle velocity at the interface (see Chapter 1). The reflection factor and absorption coefficient are related by

$$R = \frac{Z_1 - Z_0}{Z_1 + Z_0} \quad \alpha = \frac{4Z'_1 Z_0}{(Z'_1 + Z_0)^2 + Z''_1{}^2} \quad (8.37)$$

where $Z_0 = \rho_0 c_0$ is the characteristic impedance of air for plane waves and ρ_0 is the density and c_0 the speed of sound in air.

For sound absorbers consisting of highly porous layers with perforated or cloth facings of smaller porosity, a modified air-side surface impedance must be used in Eq. (8.37). This is obtained by taking the air-side volume velocity averaged over the face area of the absorber. If an absorber with surface impedance Z_i is covered by a perforated facing with porosity h , then $Z_1 = Z_i/h$.

If $|R| \rightarrow 0$, Eq. (8.37) indicates that $Z_1 \rightarrow Z_0$. This means that the ideal absorber should have a surface impedance similar to that of unbounded air. For a thick (effectively semi-infinite) layer, this requires that the characteristic impedance of porous sound-absorbing materials, Z_C , should be only slightly above that for air, Z_0 , and so it is imperative to keep the porosity high. For fibrous sound-absorbing materials, the porosity should be in the range of 0.95–0.99 and for perforated facings the porosity should be above 0.25.

This is illustrated in Fig. 8.10, which shows random-incidence absorption coefficients $\sigma_R(\text{rev})$ for various configurations with a glass fiber fabric or perforated plate covering measured in a reverberation room. Note the relatively

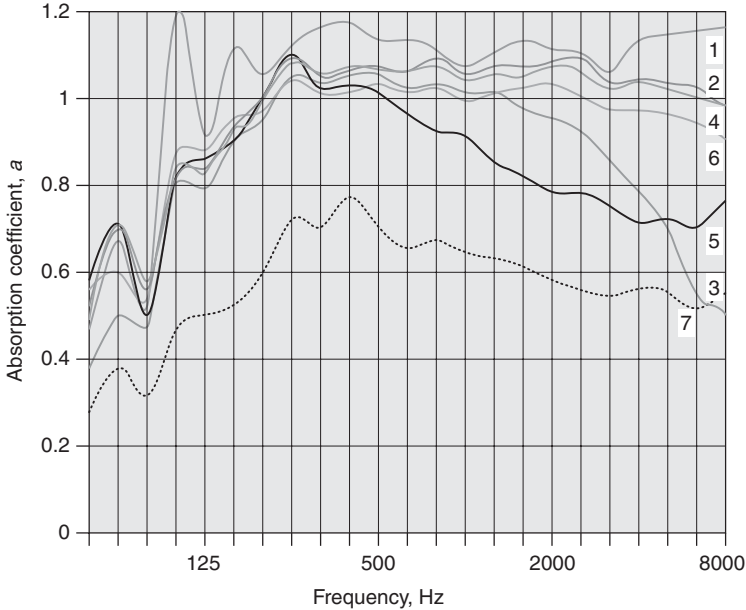


FIGURE 8.10 Random-incidence sound absorption coefficients of 200-mm-thick exhaust silencer pillows (basalt fiber core, density 125 kg/m^3) as a function of frequency measured in a reverberation room: (1) high-porosity fabric covering, equivalent flow resistivity $12 \text{ kPa} \cdot \text{s} \cdot \text{m}^{-2}$; (2) perforated sheet (32% open area) over high-porosity fabric covering; (3) perforated sheet (17% open area) over high-porosity fabric covering; (4) high-porosity fabric covering and fiberglass facing over basalt fiber core; (5) low-porosity fabric covering, equivalent flow resistivity $43 \text{ kN} \cdot \text{m} \cdot \text{s}^{-4}$; (6) perforated sheet (32% open area) and fiberglass facing over basalt fiber core; (7) perforated sheet (17% open area), low-porosity fabric covering and fiberglass facing. (Courtesy of John Bowman, Lancaster Glass Fiber, Ltd.)

poor results for the configurations of low-percentage open-area perforated-plate (17%), low-porosity, high-flow-resistivity fabric covering ($43,000 \text{ mks rays/m} = 100 \rho_0 c_0$ per meter $= 2.4 \rho_0 c_0$ per inch). Further note that the measured values of $\alpha_R(\text{rev})$ exceed unity. The reasons for this behavior have been discussed in the introductory part of this chapter.

Normal Incidence on a Porous Layer in Front of a Rigid Wall

A rigid wall represents a nearly [see Eq. (8.4)] perfect acoustical reflector. It is an acoustically hard surface. When a porous layer is placed on such an acoustically hard backing, the surface impedance is controlled by the combination of the incident and (multiple) reflected sound waves in the layer, yielding

$$Z_1 = Z_{Ca} \coth(jk_a d) = -jZ_{Ca} \cot(k_a d) \quad \text{N} \cdot \text{s/m}^3 \quad (8.38a)$$

where d is the layer thickness and Z_{Ca} is the characteristic impedance.

Equation (8.38a) may be expressed as

$$Z_1 = Z_{Ca} \frac{\sinh(bd) \cosh(bd) - j \sin(ad) \cos(ad)}{\cosh^2(bd) - \cos^2(ad)} \quad \text{N} \cdot \text{s/m}^3 \quad (8.38b)$$

The impedance given by Eqs. (8.38) exhibits the following behavior:

- (a) When $d \ll \frac{1}{4} \lambda_a$ [$\lambda_a = 2\pi/a$ is the wavelength in the absorbing material, and a is defined in Eq. (8.30a)], which is the case when the layer is thin or the frequency is low, the magnitude of $\coth(jk_{an}d)$ is always large and the lack of impedance matching between Z_1 and Z_0 leads to a small sound absorption coefficient. This is the reason there is no “sound-absorbing paint” and that rugs absorb sound only modestly.
- (b) When d is sufficiently large compared with the wavelength inside the material and the attenuation constant is not too small, the acoustical behavior of the layer approximates that of an “infinitely thick” layer. In this case $\coth(jk_{an}d) \rightarrow 1$ and the surface impedance is the same as the characteristic impedance.
- (c) When $\coth(jk_{an}d)$ is minimum, the surface impedance of the hard-backed layer is minimum and the absorption coefficient takes its maximum value. Consideration of the expanded form in Eq. (8.38b) indicates that, for $bd > 0$, which is the only situation of interest, the real part is never zero. So the normalized surface impedance is minimum approximately when the imaginary part is zero. In fact, this condition gives a slight underestimate of the frequency of maximum absorption, whereas $\cos(ad) = 0$, corresponding to $d = \frac{1}{4} \lambda_a$, gives an overestimate.

In general the absorption coefficient is maximum at a frequency that is somewhat less than the frequency at which the layer is a quarter-wavelength thick. Doubling the layer thickness halves the frequency of maximum absorption.

For practical purposes a layer of thickness d can be considered “infinitely thick” if $bd > 2$. For such conditions $Z_1 = Z_{Ca}$. The higher the flow resistivity of the porous material, the more Z_1 exceeds Z_0 . Even in the extreme case of practically infinite layer thickness (which at low frequencies and low flow resistivity becomes very large), the sound absorption coefficient will be small. As frequency increases, the general tendency is $Z_{Ca} \rightarrow Z_0/h$, and for highly porous absorbers where $h \approx 1$, the good impedance matching results in a high absorption coefficient.

Sound-Absorbing Layer Separated from a Rigid Wall by an Air Space. The sound-absorbing configuration depicted in Fig. 8.11 is frequently used in practical applications (e.g., hung acoustical ceilings). The impedance of a layer of air of thickness t in front of a rigid wall is

$$Z_2 = -j Z_0 \cot(k_0 t) \quad \text{N} \cdot \text{s/m}^3 \quad (8.39)$$

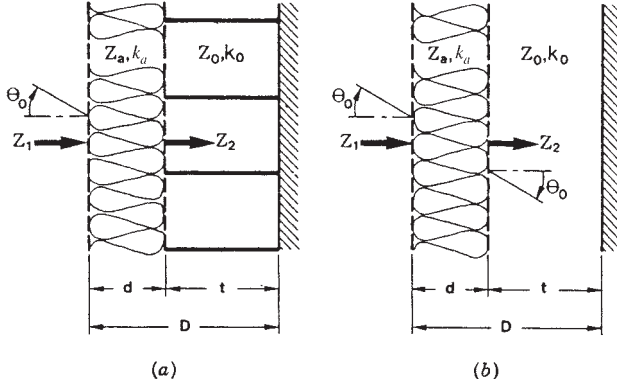


FIGURE 8.11 Combination of a bulk-reacting absorber layer with (a) an air gap such that sound can only travel perpendicular to the hard wall and (b) an air gap such that sound can travel parallel to the hard wall.

where $k_0 = \omega/c_0 = 2\pi\lambda_0/c_0$ is the acoustical wavenumber for air. The internal reflection coefficient at the back of the layer is $R_B = (Z_2 - Z_{Ca})/(Z_2 + Z_{Ca})$ and the surface impedance at the front of the layer is

$$Z_1 = Z_C \left(\frac{Z_2 + jZ_{Ca} \tan(k_a d)}{Z_{Ca} + jZ_2 \tan(k_a d)} \right) \text{ N} \cdot \text{s/m}^3 \quad (8.40a)$$

where Z_C is the normalized characteristic impedance of the porous layer, k_a is the (not-normalized) propagation constant in the porous layer, and d is the layer thickness. Note that Z_2 is zero when $t = \frac{1}{4}\lambda_0$; that is, the impedance Z_s of the absorbing layer backed by zero impedance is given by $Z_s = jZ_C \tan(k_a d) = Z_C \tanh(jk_a d)$. If the impedance of the absorbing layer backed by a hard wall [i.e., with $Z_2 \rightarrow \infty$ in Eq. (8.40a) or from Eq. (8.38a)] is denoted by Z_h , then $Z_{Ca} = \sqrt{Z_h Z_s}$. So Eq. (8.40a) can be written in the form

$$Z_1 = \frac{Z_h(Z_2 + Z_s)}{Z_2 + Z_s} = Z_h \frac{1 + Z_s/Z_2}{1 + Z_s/Z_2} \text{ N} \cdot \text{s/m}^3 \quad (8.40b)$$

The reflection factor and absorption coefficient a are computed by using Z_1 in Eq. (8.37). For small air space thickness such that $t/\lambda_0 < \frac{1}{8}$, $|Z_h| \gg |Z_w|$ and the following approximation is valid:

$$Z_1 \approx \frac{Z_h}{1 + Z_h/Z_2} \text{ N} \cdot \text{s/m}^3 \quad (8.41)$$

When the thickness of the absorbing layer is not too small, that is, $|Z_h| \ll |Z_2|$, Eq. (8.41) results in $Z_1 \approx Z_h$. Consequently at low frequencies a very thin air space between the absorbing layer and the rigid wall is ineffective.

As long as the layer thickness is small compared with the wavelength ($d < \frac{1}{8}\lambda_a$), the air space behind the absorbing layer results in a reduction of the surface impedance and at low frequencies the magnitude of the surface impedance shifts toward the characteristic impedance of air Z_0 . This results in a decrease of the reflection factor R and a corresponding increase of the normal-incidence sound absorption coefficient. The largest improvement is observed when the thickness of the air space is a quarter of a wavelength, $t = \frac{1}{4}\lambda_0$.

When the air space thickness corresponds to a multiple of $\frac{1}{2}\lambda_0$, $\sin(k_0 t) \rightarrow 0$ and Z_2 is very large [see Eq. (8.39)]. At such frequencies, the air space becomes totally ineffective.

Oblique Sound Incidence

For oblique sound incidence one must distinguish between locally reacting and bulk-reacting absorbers. Locally reacting absorbers are those where sound propagation parallel to the absorber surface is prohibited, as, for example, in partitioned porous layers, porous layers backed by a partitioned air space as depicted on the left in Fig. 8.1, in Helmholtz resonators with partitioned volumes, and in small plate absorbers. Local reaction is a good approximation to absorber behavior also when the flow resistivity is relatively high, causing the transmitted wave to bend toward the normal to the surface. This is true, for example, at high densities or in granular media consisting of small particles. In bulk-reacting absorbers, such as a low-flow-resistivity porous layer, possibly with an unpartitioned air space behind it (Fig. 8.11b), sound propagation in the direction parallel to the absorber surface is possible in the sound-absorbing layer or in the air space behind it. The term “locally reacting” results from the fact that the particle velocity at the interface depends only on the *local* sound pressure. For bulk-reacting absorbers, the particle velocity at the interface depends not only on the local sound pressure but also on the particular distribution of the sound pressure in the entire absorber volume. Certain materials are inherently anisotropic; that is, the properties vary with direction through the material. This is true, for example, of fibrous materials where the fibers lie in planes parallel to the surface. Strictly, in such materials the propagation constant and characteristic impedance are a function of angle. However, this complication will not be considered further here.

Oblique-Incidence Sound on Locally Reacting Absorber. Locally reacting absorbers are characterized by a surface impedance Z_1 that is independent of the angle of incidence θ_0 . The reflection factor $R(\theta_0)$ for locally reacting absorbers is determined by how well the surface impedance Z_1 matches the field impedance of $Z_0/\cos \theta_0$ and is given by

$$R(\theta_0) = \frac{Z_1 \cos \theta_0 - Z_0}{Z_1 \cos \theta_0 + Z_0} \quad (8.42)$$

Equation (8.42) indicates that best impedance match, and correspondingly the lowest reflection factor for a given incidence angle θ_0 , is achieved by $|Z_1| > Z_0$

and that such “overmatched” surface impedances will yield absorption maxima at a specific angle of incidence.

Oblique-Incidence Sound on Bulk-Reacting Absorber. Mineral-fiber and open-pore foams are the most frequently used bulk-reacting sound-absorbing materials. As long as the materials may be treated as isotropic, the acoustical behavior of these materials can be fully characterized by their propagation constant k_a and characteristic acoustical impedance Z_a . Methods for measuring and predicting these key acoustical parameters are given in Section 8.4 under Physical Characteristics of Porous Materials and their Measurement.

Semi-Infinite Layer. Figure 8.12 shows the incident, reflected, and transmitted waves at the interface. The combination of these waves must satisfy the following boundary conditions: (1) equal normal components of the impedances and (2) equality of the wavenumber components parallel to the interface. The second of these requirements yields the refraction law

$$\frac{\sin \theta_1}{\sin \theta_0} = \frac{k_0}{k_a} \quad (8.43)$$

where θ_1 is the complex propagation angle of the sound inside the absorber. The complex reflection factor is given by

$$R(\theta_0) = \frac{1 - (Z_0/Z_1) \cos \theta_0}{1 + (Z_0/Z_1) \cos \theta_0} \quad (8.44a)$$

and

$$Z_1 = \frac{Z_{Ca}}{[1 - (k_0/k_a)^2 \sin^2 \theta_0]^{1/2}} \quad \text{N} \cdot \text{s/m}^3 \quad (8.44b)$$

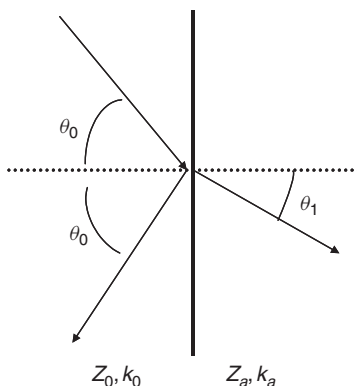


FIGURE 8.12 Reflection and transmission of an oblique-incidence sound wave by a semi-infinite, bulk-reacting absorber.

Finite-Layer Thickness. For a finite layer of bulk-absorbing material of thickness d in front of a hard wall, the reflection factor is

$$R(\theta_0) = \frac{1 - (Z_0/Z_{1d})/\cos \theta_0}{1 + (Z_0/Z_{1d})\cos \theta_0} \quad (8.45)$$

where $Z_{1d} = (Z_{Ca}/\cos \theta_1) \coth(jk_a d \cos \theta_1)$ and θ_1 is defined in Eq. (8.43),

Multiple Layers. Absorbers may consist of a porous sound-absorbing layer with an air space behind, as illustrated in Fig. 8.11, or a number of porous layers of different thicknesses and different acoustical characteristics (Fig. 8.13). Analytical expressions describing the absorber functions and design charts to predict the sound absorption coefficients for those multiple-layer absorbers are provided in reference 46.

Random Incidence. In a diffuse sound field where the intensity $I(\theta) = I$ is independent of the incident angle θ , the sound power incident on a small surface area dS of an absorber is given as

$$dW_{\text{inc}} = I dS \int_0^{2\pi} d\phi \int_0^{\pi/2} \cos \theta \sin \theta d\theta = \pi I dS \quad (8.46)$$

and the absorbed portion as

$$\begin{aligned} dW_a &= I ds \int_0^{2\pi} d\phi \int_0^{\pi/2} \alpha(\theta) \cos \theta \sin \theta d\theta \\ &= 2\pi I dS \int_0^{\pi/2} \alpha(\theta) \cos \theta \sin \theta d\theta \end{aligned} \quad (8.47)$$

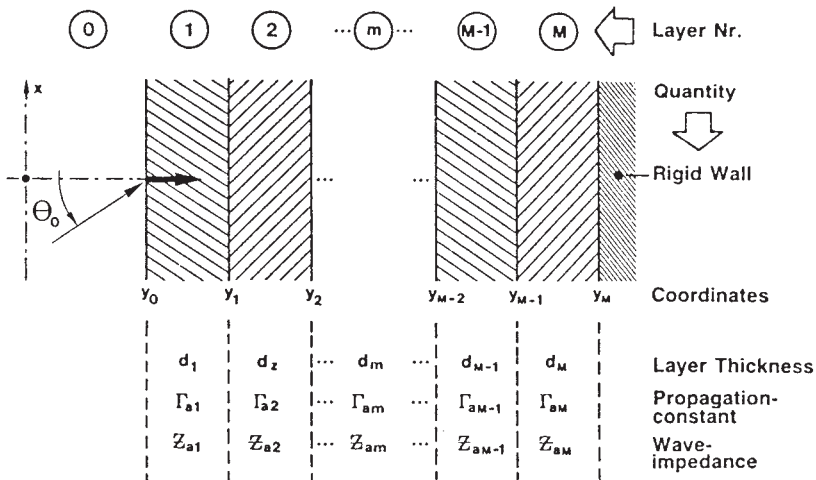


FIGURE 8.13 Multiple-layer absorber.

So the random-incidence sound absorption coefficient is given by

$$\alpha_R = \frac{dW_a}{dW_{inc}} = 2 \int_0^{\pi/2} \alpha(\theta) \cos \theta \sin \theta d\theta \quad (8.48)$$

Random-Incidence Sound Absorption Coefficient of Locally Reacting Absorbers. When the constituent materials of a multiple-layer absorber are locally reacting, the combination of Eqs. (8.2), (8.42), and (8.48) will yield the random-incidence sound absorption coefficient. The surface impedance Z_1 that is required for the computation of the random-incidence absorption coefficient can be measured in an impedance tube or computed, and the true random-incidence sound absorption coefficient α_R can be determined as⁴

$$\alpha_R = \frac{8}{|z|} \cos(\beta) \left\{ 1 + \frac{\cos 2\beta}{\sin \beta} \arctan \left(\frac{|z| \sin \beta}{1 + |z| \cos \beta} \right) - \frac{\cos \beta}{|z|} \ln(1 + 2|z| \cos \beta + |z|^2) \right\} \quad (8.49)$$

where $z = Z/\rho_0 c_0$ is the normalized wall impedance and $\beta = \arctan[\text{Im}(z)/\text{Re}(z)]$.

The maximum value of the true $\alpha_R(\text{max}) = 0.955$ that occurs at a normalized wall impedance of $|z| = 1.6$ and $\beta \approx \pm 30^\circ$ indicates that no locally reacting sound absorber can achieve total absorption or even the 99% absorption that is required as a minimum for the walls of anechoic chambers. The random-incidence sound absorption coefficient measured in a reverberation room, $\alpha_R(\text{rev})$, as described in Chapter 7, can yield physically impossible values above unity (sometimes in excess of 1.2). To avoid confusion, the computed physically correct random-incidence coefficient represented by α_R and that measured with the reverberation room method should have a different symbol, namely $\alpha_R(\text{rev})$.

Random-Incidence Sound Absorption Coefficient of Bulk-Reacting Absorbers. For bulk-reacting absorbers the integration in Eq. (8.48) can only be evaluated numerically. It is more appropriate to use the geometric and acoustical characteristics of the porous liner than its impedance to generate design charts for general use.

8.6 DESIGN CHARTS FOR FIBROUS SOUND-ABSORBING LAYERS

For design use it is useful to have charts where the sound absorption coefficient is plotted as a function of the first-order parameters of the absorber, such as thickness d , flow resistivity R_1 , and frequency f . This section contains such design charts. Design charts for a multiple-layered absorber, such as shown in Fig. 8.13, are given in reference 46.

Monolayer Absorbers

As shown in Fig. 8.7*b*, the key acoustical parameters k_a and Z_{Ca} of fibrous sound-absorbing materials depend only on a single material parameter, namely the flow resistivity R_1 (specific flow resistance per unit thickness). Consequently, the absorption coefficient can be computed and plotted as a function of two dimensionless variables:

$$F = \frac{d}{\lambda} = \frac{fd}{c_0} \quad \text{and} \quad R = \frac{R_1 d}{Z_0} \quad (8.50)$$

Then $k_0 d = 2\pi F$ and $E = \rho_0 f / R_1 = F/R$. At oblique incidence, the angle θ is the third input variable. In the design charts contour lines of constant values of α are plotted on a log-log chart of R versus F . A variation of frequency f produces a horizontal path, an increase of the flow resistivity R_1 causes an upward move, and an increase in layer thickness d leads to a diagonally upward shift to the right. Frequency curves for an absorber layer may be derived by a horizontal intersection, with a starting point determined by the individual parameter values.

Figure 8.14 shows lines of constant absorption of a monolayer absorber for *normal incidence* (the absorber may be bulk or locally reacting). The maximum absorption is reached at about $R = 1.2$ for $F = 0.25$, that is, at a thickness d equal to a quarter of a free-field wavelength and at the flow resistance of the layer of $1.2Z_0$. Higher resonances at odd multiples of a quarter wavelength are visible for a small flow resistance. The “summit line” of the first maximum, which goes through the points of relative maxima of α at the first resonance, is inclined so

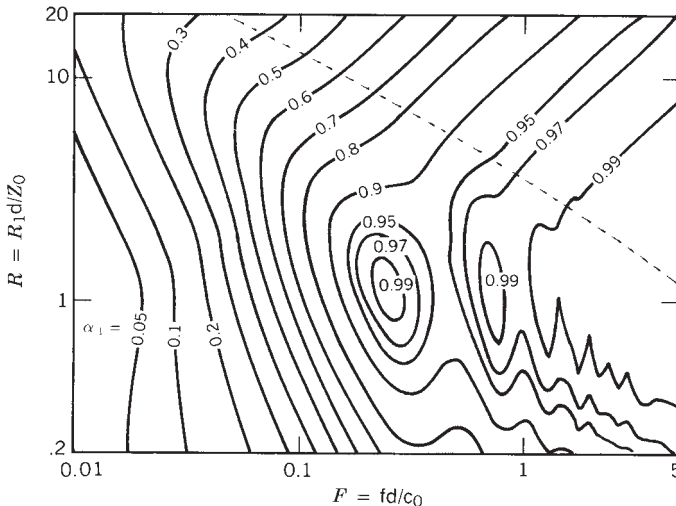


FIGURE 8.14 Lines of constant absorption at normal-incidence sound; absorption coefficient for fibrous absorbers of thickness d and flow resistivity R_1 ; no air space.

that the first relative maximum will occur below $F = 0.25$ for flow resistances such that $R > 1$, and the first relative maximum of the frequency curve will occur at somewhat higher frequencies than $F = 0.25$ for flow resistances such that $R < 1$. The higher orders of resonances are relatively stationary in frequency as R varies.

At *oblique incidence* the sound absorption for small angles, below about $\theta = 30^\circ$, is only modified slightly. The modifications in $\alpha(\theta)$ are pronounced for larger angles with locally reacting absorbers. The curves on the left in Figs. 8.15 *a–c* were computed for a locally reacting absorber for θ values of 30° , 45° , and 60° . The resonance structure of the contour plots becomes very distinct at low flow resistances and high frequencies.

Curves on the right in Figs. 8.15 *a–c* show the analogous contour plots of $\alpha(\theta)$ for the same values of θ but now for bulk-reacting (isotropic) absorber layers. Compared to those for a local-reacting absorber, the contours are relatively continuous. The maxima of absorption are shifted downward toward smaller R values for larger angles.

The *random-incidence* absorption coefficients α_R for locally reacting and bulk-reacting absorbers are plotted on the left and right, respectively, in Fig. 8.16. For locally reacting absorbers the absolute maximum of $\alpha_R = 0.95$ is attained for a matched flow resistance $R = 1$ at $F = 0.367$, that is, at a thickness $d = 0.367\lambda_0$, which is larger than a free-field quarter wavelength $\frac{1}{4}\lambda_0$. With homogeneous isotropic absorbers, the absorption coefficient α_R becomes larger than with locally reacting absorbers. Only a weak resonance maximum (belonging to the second resonance) is observed with homogeneous absorbers. The contour lines of absorption are smooth and steady.

Key Information Contained in Design Charts. Many of the general conclusions that were known qualitatively can be answered now in a quantitative manner by studying Figs. 8.14–8.16.

Determining Thickness of Absorber. What is the thickness d_∞ from which an absorber layer starts to behave acoustically as infinitely thick and yields no further increase in absorption? This thickness starts where the contour lines become 45° diagonally straight lines. The exact position of the curve for thickness d_∞ will depend on the criterion chosen and on the tolerance allowed. If, at this limit, the final value of α is supposed to be reached with a deviation between 1 and 3%, then the limit curve for normal incidence (in Fig. 8.14) is defined by

$$F = 7.45R^{-1.67} \quad (8.51)$$

The layer becomes practically infinite when the sound attenuation during propagation of the sound wave through the layer from the surface to the rigid rear wall is $8.68\Gamma'_a d_\infty = 24$ dB, as first derived in reference 2. This is a much higher attenuation than usually assumed as sufficient (about 6–10 dB) for neglecting the influence of the reflection from the rigid wall on the sound absorption at the front side of the absorber layer.

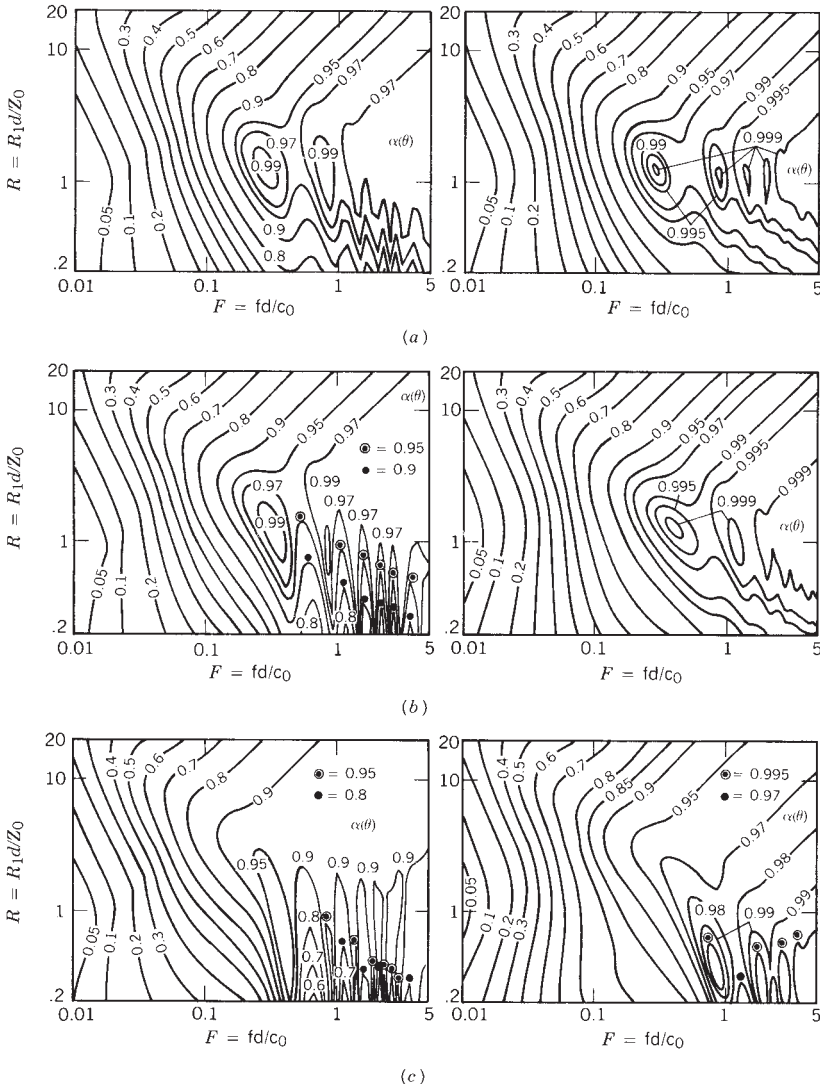


FIGURE 8.15 Lines of constant absorption $\alpha(\theta)$ for fibrous absorbers of thickness d and flow resistivity R_1 for discrete angles of incidences θ ; no air space. Left: locally reacting. Right: bulk reacting. (a) $\theta = 30^\circ$; (b) $\theta = 45^\circ$; (c) $\theta = 60^\circ$.

The relationships for “practically infinite” layer thickness d_∞ for which $\Delta L(d_\infty) = 24$ dB can be formulated as

$$f d_\infty^{2.67} R_1^{1.67} = 59 \times 10^6 \quad \text{for } \theta = 0^\circ \quad (8.52a)$$

$$f d_\infty^{2.56} R_1^{1.56} = 34.3 \times 10^6 \quad \text{for } \theta = 45^\circ, \\ \text{locally reacting and random incidence} \quad (8.52b)$$

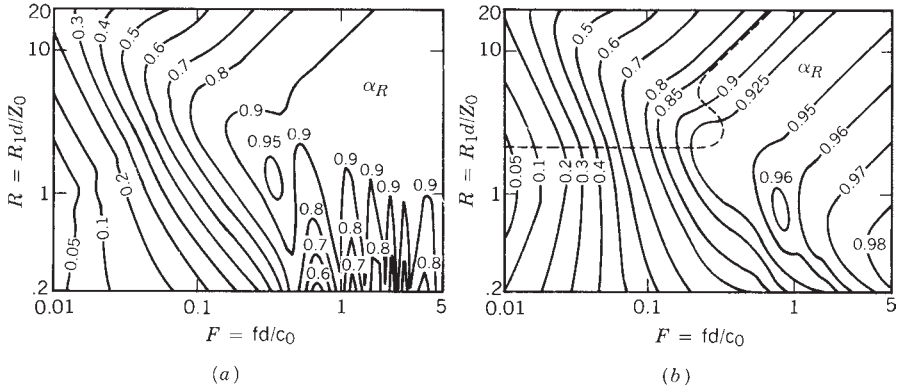


FIGURE 8.16 Lines of constant random-incidence absorption coefficient α_R for fibrous absorbers of thickness d ; no air space: (a) locally reacting liner; (b) bulk-reacting liner.

$$fd_{\infty}^{2.4} R_1^{1.4} = 7.7 \times 10^6 \quad \text{for } \theta = 45^\circ, \text{ bulk reacting} \quad (8.52c)$$

$$fd_{\infty}^{2.2} R_1^{1.2} = 1.1 \times 10^6 \quad \text{for random-incidence bulk reacting} \quad (8.52d)$$

where f = frequency, Hz

d_{∞} = layer thickness, m

R_1 = flow resistivity, $\text{N} \cdot \text{s}/\text{m}^4$

Optimal Choice of Flow Resistance. Optimal values for the normalized specific flow resistance of the layer, $R = R_1 d / Z_0$, plotted on the vertical scale of Figs. 8.14–8.16, depend on what the designer wants to accomplish.

If the aim is to maximize the sound absorption coefficient, then the optimal choice of normalized specific flow resistance R is in the range of 1–2 for both locally reacting and bulk-reacting absorbers and for normal, oblique, or random angles of sound incidence with the exception of bulk-reacting absorbers for $\theta > 45^\circ$ incidence where $R = 0.7$ yields the best results.

The absorption coefficient depends strongly on flow resistivity R_1 , where the curves in Figs. 8.14–8.16 have nearly horizontal contours. The design charts reveal that the typical orientation of the curves of constant α is nearly vertical, indicating only a slight dependence on R_1 . This is fortunate because our ability to determine R_1 and our accurate control over its value in the manufacturing process are limited.

Bulk-Reacting versus Locally Reacting Absorbers. A comparison of two graphs in Fig. 8.16 indicates that the random-incidence sound absorption coefficient of locally reacting and bulk-reacting absorbers of the same thickness is practically identical (within 10%) in the left upper quadrant, where $R > 2$ and $F = fd/c_0 < 0.25$, above the dotted line in the graph on the right in Fig. 8.16.

A comparison of $\theta = 45^\circ$ curves in Fig. 8.15b with the random-incidence curves presented in Fig. 8.16 indicates that the $\theta = 45^\circ$ curves for both locally

reacting and bulk-reacting absorbers agree with the random-incidence curves within an error of 5%, except in the range where $\alpha > 0.9$, which is of little practical interest. This close agreement between $\alpha(45^\circ)$ and α_R indicates that it is permissible to compute $\alpha(45^\circ)$ instead of the much more difficult α_R .

Two-Layer Absorbers

The two-layer absorbers shown in Fig. 8.11 have a layer of fibrous sound-absorbing material of thickness d backed by an air space of thickness t , with total thickness $D = d + t$. Figure 8.11*a* shows a partitioned (e.g., in a honeycomb pattern) air space resulting in a locally reacting absorber, and Fig. 8.11*b* shows an unpartitioned air space resulting in a bulk-reacting absorber. The sound absorber types shown in Fig. 8.11 are usually employed when it is impractical to fill the entire absorber thickness with porous material because no fibrous material of sufficiently low flow resistivity is available to keep the normalized flow resistance $R = R_1 d / Z_0 < 2$. Based on the analyses described in reference 5, design charts similar to those presented in a preceding section for monolayer absorbers were computed.

Figure 8.17 shows contour maps of the random-incidence sound absorption coefficient α_R in a diffuse sound field for a bulk absorber layer of thickness d in front of a locally reacting air gap of thickness t (see Fig. 8.11*a*) for three fractions, $d/D = 0.25, 0.5, 0.75$, absorber thickness d , and total thickness $D = d + t$. Because the two-layer design curves in Fig. 8.17 are not plotted as functions of the total layer thickness D , they are not directly comparable with the monolayer design curves presented in Fig. 8.16. Some differences may be noticed by comparing these figures with Fig. 8.16, which applies to the bulk monolayer absorber.

First, the variable $F = fd/c_0$ for maximum absorption in the first resonance maximum is shifted toward smaller values with decreasing d/D , which results in higher absorption at low frequencies for less absorber material thickness. Generally, however, the reduction in weight of the material is not as large. The horizontal shift of the absorption maximum toward lower F values, when realized by a reduction in d , makes a larger flow resistivity necessary on the ordinate in order to hold R on a constant value. The increase in R_1 is achieved mostly by an increase of the material bulk density ρ_A . For most fibrous absorber materials R_1 is proportional to about $\rho_A^{1.5}$. Hence, only a small net reduction in absorber material is possible by the addition of an air gap.

For thick absorber layers, that is, for large d , the air gap has no effect, as expected. The limit for the onset of an “infinite” thickness is the same as with the monolayer absorber.

The resonance structure of the layered absorber with small d/D resembles more that of the locally reacting monolayer absorber (see Fig. 8.16*a*). The character of the bulk-reacting monolayer absorber plot (see Fig. 8.16*b*), which must be the asymptotic limit for increasing d/D , becomes dominant for $d/D \geq 0.5$.

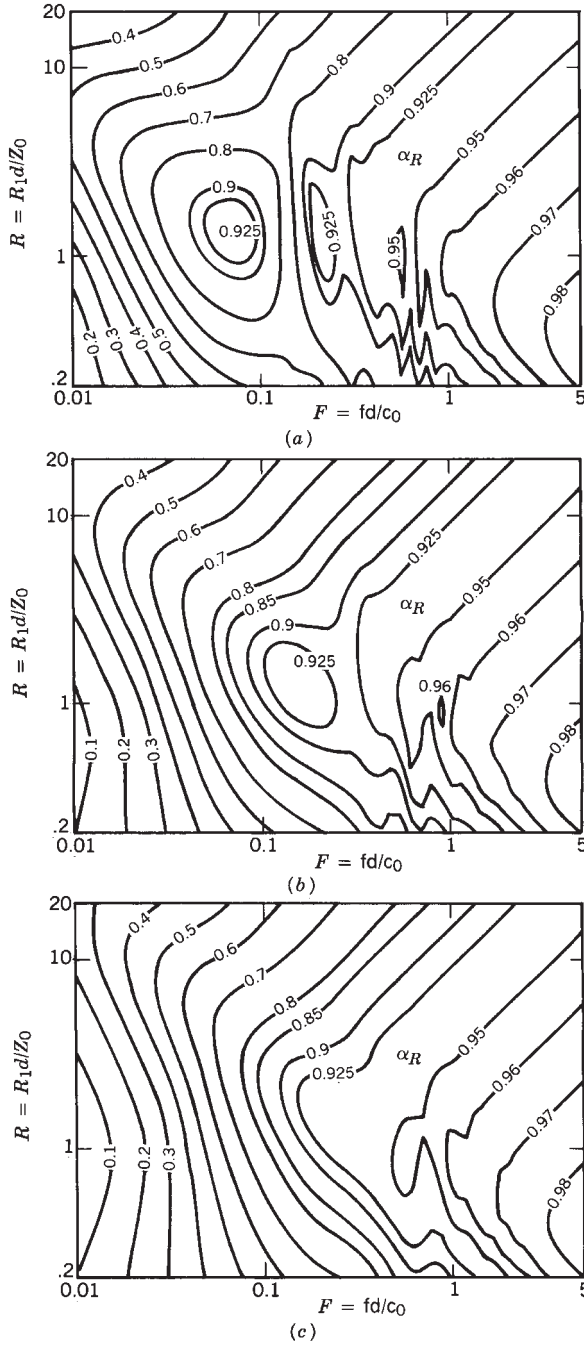


FIGURE 8.17 Lines of constant random-incidence absorption coefficient α_R for a bulk-reacting absorber layer of thickness d in front of a locally reacting air gap of thickness t for thickness ratios $d/D = d/(d + t)$ of (a) 0.25, (b) 0.5, and (c) 0.75.

The sound absorption coefficient of a porous layer of thickness d with an air gap behind it is generally higher than that of a monolayer with equal d . However, the absorption of the layered absorber is smaller than that of a monolayer absorber with equal total thickness $d' = D$. In the range $R > 1$, the absorption of a layered absorber with $d/D = 0.75$ corresponds to that of a monolayer absorber with thickness $d' = 1.25d$ and the absorption of a layered absorber with $d/D = 0.5$ corresponds to that of a monolayer having a thickness $d' = 1.67d$. The quite different character of the plot for $d/D = 0.25$ excludes a similar equivalence.

Finally, the lines of constant α_R of the layered absorber have strong deflections toward the left for values of α_R between about 0.7 and 0.9. As a consequence, the optimum flow resistance $R_1 d$ (which is defined by the leftmost point of a curve) has a more distinct meaning than with monolayer absorbers.

Design graphs for layered absorbers with bulk-reacting absorber layers in front of a bulk-reacting air gap (see Fig. 8.7*b*) are given in reference 45.

Multilayer Absorbers

Multilayer absorbers, such as shown in Fig. 8.13, have been treated in reference 4. Best results are obtained if the flow resistivity of the layers, R_1 , increases from the interface toward the rigid wall. These results obtained with such multilayer absorbers are somewhat better than those obtained with a monolayer absorber of equal thickness. However, the improvements seldom justify the added cost and complexity. Of course, Fig. 8.13 represents an idealization of practical constructions (see Fig. 8.18) which combine multiple porous layers with perforations and slots.

It should be noted, however, that even the most elaborate multiple-layer absorber cannot match the random-incidence sound absorption performance of anechoic wedges.

Thin Porous Surface Layers

Thin porous surface layers such as mineral wool felt sprayed on plastic, steel wool, mineral wool, or glass fiber cloth; wire mesh cloth; and thin perforated metal are frequently used to provide mechanical protection. They also reduce the

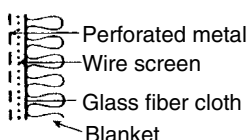


FIGURE 8.18 Multilayered absorber with perforated porous layer structure.

loss of fibers. The acoustical properties of such thin layers are characterized by their flow resistance $R_s = \Delta p/v$ and their mass per unit area ρ_s . The surface impedance is

$$Z_s = \begin{cases} R_s & \text{(fixed)} \\ \frac{(j\omega\rho_s)R_s}{j\omega\rho_s + R_s} & \text{(free)} \end{cases}$$

(8.53a)

(8.53b)

which must be added to the wall impedance of the absorber. If the porous surface layer is not free to move, Eq. (8.53a) is used. For surface layers that are free to move due to the pressure differential produced by their flow resistance, Eq. (8.53b) is used.

Tables 8.5 and 8.6 provide design information on wire mesh cloth and glass fiber cloth, respectively. The flow resistance values listed in these tables represent the linear part of the flow resistance, which is appropriate for design use only if the particle velocity is low. At high sound pressure levels (above 140 dB), the nonlinear behavior of the flow resistance must be determined by measuring the flow resistance as a function of the face velocity.

Sintered porous metals have been developed for silencers in jet engine inlets. If backed by a honeycomb-partitioned air space, they provide good sound absorption and remain linear up to high sound pressure levels and for high-Mach-number grazing flow. Their additional advantage is that they do not require any surface protection. Table 8.7 gives the flow resistance R_s and mass per unit area ρ_s for some commercially available porous metal sheets.

Steel wool mats typically have a thickness from 10 to 45 mm, mass per unit area ρ_s from 1 to 3.8 kg/m², and a specific flow resistance R_s from 100 to 500 N · s/m³. Recently, such steel wool mats needled on mineral wool of specified thickness and density have become available on special order.

Perforated metal facings, when they rest directly on a porous sound-absorbing layer, can be accounted for by a series impedance Z_s given by

$$Z_s = \frac{j\omega\rho_0}{\varepsilon} \left\{ l + \Delta l \left[H(1 - |v_h|) - j \frac{\Gamma_a}{k_0} \frac{Z_a}{Z_0} \right] \right\} \tag{8.54}$$

TABLE 8.5 Mechanical Characteristics and Flow Resistance R_s of Wire Mesh Cloths

Wires/cm	Wires/in.	Wire Diameter		Mass per Unit Area		Flow Resistance R_s	
		μm (10^{-6} m)	mils (10^{-3} in.)	kg/m^2	lb/ft^2	$\text{N} \cdot \text{s/m}^3$	$\rho_0 c_0$
12	30	330	13.0	1.6	0.32	5.7	0.014
20	50	220	8.7	1.2	0.25	5.9	0.014
40	100	115	4.5	0.63	0.13	9.0	0.022
47	120	90	3.6	0.48	0.1	13.5	0.033
80	200	57	2.25	0.31	0.63	24.6	0.06

TABLE 8.6 Mechanical Characteristics and Flow Resistance R_s of Glass Fiber Cloth

Manufacturer ^b	Cloth Number	Surface Density ^a		Construction, Ends $\times$ Picks	Flow Resistance, mks rayls ($N \cdot s/m^3$)
		oz/yd ²	g/m ²		
1, 2, 3	120	3.16	96	60 $\times$ 58	300
1, 2, 3	126	5.37	164	34 $\times$ 32	45
1, 2, 3	138	6.70	204	64 $\times$ 60	2200
1, 2, 3	181	8.90	272	57 $\times$ 54	380
3	1044	19.2	585	14 $\times$ 14	36
2	1544	17.7	535	14 $\times$ 14	19
3	3862	12.3	375	20 $\times$ 38	350
1	1658	1.87	57	24 $\times$ 24	10
1	1562	1.94	59	30 $\times$ 16	<5
1	1500	9.60	293	16 $\times$ 14	13
1	1582	14.5	442	60 $\times$ 56	400
1	1584	24.6	750	42 $\times$ 36	200
1	1589	12.0	366	13 $\times$ 12	11

^a Averaged over a large sample.^b Code numbers for manufacturers are as follows: (1) Burlington Glass Fabrics Company; (2) J. P. Schwebel and Company; (3) United Merchants Industrial Fabrics.**TABLE 8.7 Specific (Unit-Area) Flow Resistance R_s , Thickness, and Mass per Unit Area of Sintered Porous Metals Manufactured by the Brunswick Corporation^a**

Specific Flow Resistance Air, 70°F		NLF ^b	Designation	Thickness		Mass per Unit Area	
$\rho_0 c_0$	$N \cdot s/m^3$			mm	in.	kg/m ²	lb/ft ²
0.25	100	3.6	FM 125	1.0	0.04	3.9	0.79
		5.0	FM 127	0.76	0.03	3.3	0.67
		2.6	FM 185	0.5	0.02	2.0	0.4
		2.0	347-10-20-AC3A-A	0.5	0.02	1.32	0.27
		2.0	347-10-30-AC3A-A	0.76	0.03	1.1	0.23
		2.0	FM 802	0.5	0.02	1.3	0.27
		4.7	FM 134	0.89	0.035	3.8	0.77
0.88	350	1.8	FM 122	0.76	0.03	1.4	0.28
1.25	500	3.6	FM 126	0.66	0.026	3.7	0.76
		3.3	FM 190	0.41	0.016	2.0	0.4
		2.0	347-50-30-AC3A-A	0.76	0.03	1.4	0.29

^a Except FM 802, which is made of Hastelloy X, all materials are type 347 stainless steel. (Courtesy of Brunswick Corporation.)^b Nonlinearity factor, calculated as the ratio of flow resistances obtained at flow velocities of 500 and 20 cm/s, respectively.

where ε is the fractional open area of the perforated facing, l is the plate thickness, Δl is the end correction length of the perforations given in Section 8.5, H is a step function that is unity for $|v_h| \leq 1$ m/s and zero for $|v_h| > 1$ m/s, and v_h is the particle velocity in the holes of the perforations. For thin, unrestrained perforated surface protection plates, it is necessary to take the parallel combination of Z_s and $j\omega\rho_s$ according to Eq. (8.53b).

8.7 RESONANCE ABSORBERS

In building acoustics the most frequently used type of resonant absorber is the Helmholtz resonator, consisting of the mass of an air volume in a cross-sectional area restriction (such as holes or slits in a covering plate) and the compliance of the air volume behind the covering plate. In the narrow frequency range, centered at their resonance frequency, the resonators “soak in” the acoustical energy from a large room volume in the vicinity of the neck. This high energy concentration results in high local sound pressures at the neck (and in the internal volume) which are much higher than that one would register with the resonator opening closed. On the room side, this near-field sound pressure decays exponentially with increasing distance. In architectural application of resonance absorbers care should be exercised not to locate them too near to any member of the audience to avoid distortion of the auditory experience. The open bottles embedded in the vertical portion of the seats in ancient Greek amphitheaters indicate that this local amplification of the sound field in the vicinity of acoustical resonators was known more than 2000 years ago. What was not known, which misguided their application, was that the amplification occurs only in a narrow frequency band and cannot locally amplify such broadband signals as speech. Such a Helmholtz resonator (named after the nineteenth-century German physicist Ludwig Helmholtz, who was first to utilize its narrow-band tuning to analyze the spectral composition of complex sounds) is shown in Fig. 8.19. In the following treatment it is assumed that all dimensions of a single resonator are small compared with the acoustical wavelength (except in the case of two-dimensional resonators with slits) and that the skeleton of the resonator is rigid.

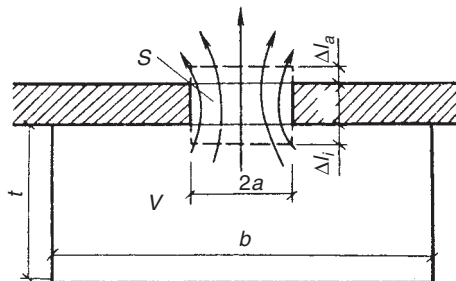


FIGURE 8.19 Key geometric parameters of a Helmholtz resonator.

Acoustical Impedance of Resonators

The specific acoustical impedance of the resonator opening Z_R is the sum of the impedance of the enclosed air volume Z_v and that of the air volume that oscillates in and around the resonator mouth Z_m , namely,

$$Z_R = Z_v + Z_m = (Z'_v + jZ''_v) + (Z'_m + jZ''_m) \quad \text{N} \cdot \text{s/m}^3 \quad (8.55)$$

The volume impedance $Z_v = jZ''_v$ is purely imaginary and predominantly of spring character while the mouth impedance has a real part Z'_m and an imaginary part Z''_m and is predominantly of mass character.

The impedance of a rectangular resonator volume is

$$Z_v = jZ''_v = -j\rho_0 c_0 \cot(k_0 t) \frac{S_a}{S_b} \quad \text{N} \cdot \text{s/m}^3 \quad (8.56a)$$

where S_b is the surface area of the resonator cover plate, $S_a = \pi a^2$ is the area of the resonator mouth, t is the depth of the resonator cavity, and $V = S_b t$ is the resonator volume. If the resonator dimensions are small compared with the wavelength ($k_0 t \ll 1$), Eq. (8.56a) yields

$$Z_v = jZ''_v = -j \frac{\rho_0 c_0^2}{\omega} \frac{S_a}{V} \quad \text{N} \cdot \text{s/m}^3 \quad (8.56b)$$

Thus, when all the dimensions are smaller than the wavelength of interest, the shape of the air cavity does not matter.

The impedance of the resonator mouth Z_m consists of components related to the oscillations of air internal to the mouth, within the mouth, and external to the mouth of the resonator plus the impedance of the screen (Z_s) that may be placed across the resonator mouth to provide resistance. For circular resonator openings of radius a and orifice plate thickness l , the resulting total impedance is given by

$$Z'_m = \rho_0 \left[\sqrt{8\nu\omega} \left(1 + \frac{l}{2a} \right) + \frac{(2\omega a)^2}{16c_0} \right] + Z_s \quad \text{N} \cdot \text{s/m}^3 \quad (8.57a)$$

$$Z''_m = \omega\rho_0 \left[l + \left(\frac{8}{3\pi} \right) 2a \right] + \left(l + \frac{1}{2a} \right) \sqrt{8\nu\omega} \quad \text{N} \cdot \text{s/m}^3 \quad (8.57b)$$

where ν is the kinematic viscosity ($\nu = 1.5 \times 10^{-6} \text{ m}^2/\text{s}$ for air at room temperature). The quantity $0.5(8\nu/\omega)^{1/2}$ is the viscous boundary layer thickness.¹

In general, the first term in Eq. (8.57a) applies only for smooth orifice edges. For sharp orifice edges it can be many times higher. Also note that the second term in Eq. (8.57b) is usually small compared with the first term and can be neglected.

The results of Eqs. (8.57) for circular orifices and for a single resonator can be generalized for other orifice shapes and for situations where the orifice is located

on a large wall or at a two- or three-dimensional corner. This generalized form is

$$Z'_m = \frac{P(l + P/2\pi)}{4S_a} \rho_0 \sqrt{8\nu\omega} + \rho_0 c_0 \frac{k_0^2 S_a / \Omega}{1 + k_0^2 S_a / \Omega} + Z_s \quad \text{N} \cdot \text{s/m}^3 \quad (8.58a)$$

$$Z''_m = j\omega\rho_0 \left[l + 2\Delta l + \frac{P(l + P/2\pi)}{4S_a} \sqrt{8\nu/\omega} \right] \quad \text{N} \cdot \text{s/m}^3 \quad (8.58b)$$

where P is the perimeter of the orifice, S_a is its surface area, and Ω is the spatial angle the resonator “looks into”:

$$\Omega = \begin{cases} 4\pi & \text{for resonator away from all walls} \\ 2\pi & \text{flush mounted on wall far from corners} \\ \pi & \text{flush mounted on wall at two-dimensional corners} \\ \frac{1}{2}\pi & \text{flush mounted on wall at three-dimensional corners} \end{cases}$$

The quantity $\Delta l = 16a/3\pi$ represents the combined internal and external end corrections.

Resonance Frequency

The resonance frequency f_0 of the Helmholtz resonator occurs where Z''_m and Z''_v are equal in magnitude. Combination of Eqs. (8.56b) and (8.57b) yields

$$f_0 = \frac{c_0}{2\pi} \sqrt{\frac{S_a}{V \langle \dots \rangle}} \cong \frac{c_0}{2\pi} \sqrt{\frac{S_a}{V(l + 16a/3\pi)}} \quad (8.59)$$

where $\langle \dots \rangle = l + \Delta l$ represents the quantity in square brackets in Eq. (8.58b).

The viscous terms in (8.57b) are ignored in (8.59). However, this is usual when calculating a resonance frequency because the viscous contribution is small. For resonators where the area of the mouth, S_a , is not much smaller than the area of the top plate of the resonator, the proximity of the side walls of the cavity influences the flow in the resonator mouth. In this case it is necessary to determine the combined end correction from $\Delta l = 8a/3\pi + \Delta l_{\text{int.}}$ ⁴⁷

Absorption Cross Section of Individual Resonators

For individual resonators (or groups of resonators where the distance between the individual resonators is large enough that interaction is negligible), the absorption coefficient α is meaningless. In this case, the sound-absorbing performance must be characterized by the absorption cross section A of the individual resonator. The absorption cross section is defined as that surface area (perpendicular to the direction of sound incidence) through which, in the undisturbed sound wave (resonator not present), the same sound power would flow through as the sound power absorbed by the resonator.

The power dissipated in the mouth of a resonator is

$$W_m = 0.5|v_a|^2 S_a R_T = \frac{0.5 \times 2^n |p_{\text{inc}}|^2 S_a R_T}{|Z_R|^2} \quad \text{W} \quad (8.60)$$

Accordingly, the absorption cross section is

$$A = \frac{2^n \rho_0 c_0 S_a R_T}{|Z_R|^2} \quad \text{m}^2 \quad (8.61)$$

where $n = 0$ for resonators placed in free space, $n = 1$ if the resonators are flush mounted in a wall, $n = 2$ if the resonators are at the junction of two planes, and $n = 3$ if the resonators are in a corner; $R_T = Z'_m + Z_{\text{rad}}$ is the total resistance.

At the resonance frequency f_0 , where $Z''_v + Z''_m = 0$, the absorption cross section reaches its maximum value A_0 :

$$A = 2^n \frac{S_a \rho_0 c_0}{R_{\text{rad}}} \left[\frac{R_T / R_{\text{rad}}}{|1 + R_T / R_{\text{rad}}|^2} \right] \quad \text{m}^2 \quad (8.62a)$$

For a flush-mounted ($n = 1$) circular resonator area $S_a = \pi a^2$, the radiation resistance is $R_{\text{rad}} = 2(\pi a)^2 \rho_0 c_0 / \lambda_0^2$, and Eq. (8.62a) yields

$$A = \frac{1}{\pi} \lambda_0^2 \left[\frac{R_T / R_{\text{rad}}}{|1 + R_T / R_{\text{rad}}|^2} \right] \quad \text{m}^2 \quad (8.62b)$$

where λ_0 is the wavelength at the resonance frequency of the resonator. The maximum value of the absorption cross section is obtained by matching the *internal* resistance to the radiation resistance at the resonance frequency, yielding ($R_T = R_{\text{rad}}$)

$$A_0^{\text{max}} = \frac{\lambda_0^2}{4\pi} \quad \text{m}^2 \quad (8.63)$$

According to Eq. (8.63), a matched resonator tuned to 100 Hz can achieve absorption cross section $A_0^{\text{max}} = 0.92 \text{ m}^2$.

According to Eq. (8.61), the frequency dependence of A can be presented in generalized form as

$$\frac{A(f)}{A_0} = \frac{1}{1 + Q^2 \phi^2} \quad (8.64)$$

where the Q of the resonator and the normalized frequency parameter ϕ are given as

$$Q = \frac{Z''_m(\omega_0)}{Z'_m(\omega_0)} \quad \phi = \frac{f}{f_0} - \frac{f_0}{f}$$

Figure 8.20 shows the normalized absorption cross section A/A_0 as a function of the normalized frequency ϕ with Q as the parameter, indicating that the

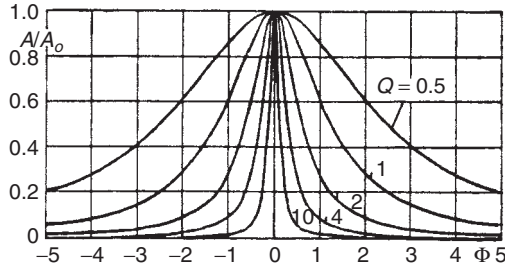


FIGURE 8.20 Normalized absorption cross section A/A_0 as function of normalized frequency ϕ with Q as parameter.

normalized bandwidth of the absorption cross section f_0/Q corresponds to the relative bandwidth $\Delta f/f_0 = 1/Q$.

Consequently, the choice of Q determines the relative shape of the absorption curve and the factor in the square brackets in Eq. (8.62b) determines the height of the curve at resonance. To obtain optimal absorption characterized by high absorption at resonance A_0 and wide bandwidth requires resistance matching $R_T = R_{\text{rad}}$ and a relatively low value of Q .

Nonlinearity and Grazing Flow

The oscillating air mass (that includes the end-correction term) is a measure for the reversible kinetic energy of the acoustical resonator. Reversibility and spatial coherence diminish with increasing turbulence due to jet flow through the orifice caused by high-amplitude sound and by the grazing flow of velocity U_∞ . Both reduce the air mass that participates in the oscillating motion and consequently increase resonance frequency and increase the losses of the resonator.

Nonlinearity and grazing flow affect only the mouth impedance Z_m of the resonator orifice. Table 8.8 is a compilation of the nonlinear effects owing to high sound pressure level and grazing flow for a perforated plate. The round holes are regularly distributed. The porosity of the perforated plate is $\varepsilon = \pi a^2/b^2$, where a is the hole radius and b the hole spacing. The amplitude nonlinearity is characterized by the particle-velocity-based Mach number $M_0 = v/c_0$, which is the ratio of the particle velocity in the resonator orifice and the speed of sound in air. The grazing flow nonlinearity is characterized by the flow-velocity-based Mach number $M_\infty = U_\infty/c_0$, where U_∞ is the velocity of the grazing flow far from the wall. The formulas presented in Table 8.8 are analytical results adjusted to yield agreement with measured data.⁴⁸ The impedances in Table 8.8 are values averaged over the plate surface.

Internal Resistance of Resonators

The most difficult part of resonator design is the prediction of the internal resistance. In the earlier section Acoustical Impedance of Resonators (page 265)

formulas were given to predict the friction loss resistance [first term in Eq. (8.57a)], which is valid only for rounded orifice perimeter. Additional resistance due to sharp edges cannot be predicted analytically. Table 8.8 contains formulas to predict resonator resistance due to nonlinear effects owing to high sound pressure level and grazing flow.

Should it be desirable to obtain higher loss resistance than provided by friction and nonlinear effects (e.g., obtaining larger absorption bandwidth), porous materials such as a screen, felt, or layer of fibrous sound-absorbing material must be placed in (or behind) the resonator orifice. Figure 8.21 shows some of the more frequently used ways to increase resonator resistance. In Fig. 8.21, R_1 is the flow resistivity of the porous material, d is its thickness, t_1 is its distance from the back side of the cover plate, t is the depth of the resonator cavity, $2a$ is the diameter of the holes, Δl is the end correction, and ε is the porosity of the cover plate.

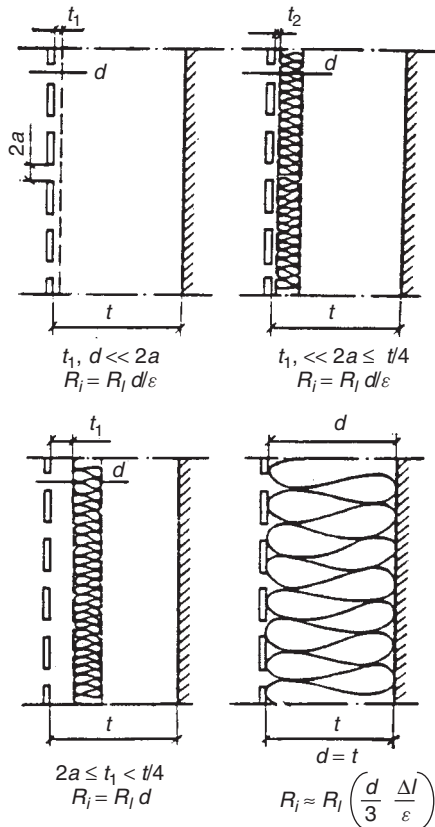


FIGURE 8.21 Methods to increase resonator resistance and the achieved specific acoustical resistance R_i (R_1 = flow resistivity of porous material, d is its thickness, and $\Delta l = 16a/3\pi$).

TABLE 8.8 Nonlinearity of Orifice Impedance $Z_m = Z'_m + jZ''_m$ of Perforated-Plate Resonator Covers Caused by High Sound Pressure Level L_p and Grazing Flow U_∞ ^a

Grazing Flow	Low SPL, $L_p < L_{0l}$	Medium SPL, $L_{0l} \leq L_p \leq L_{0h}$	High SPL, $L_p > L_{0h}$
No flow or very low flow, $M_\infty \leq 0.025$, $U_\infty \leq 8$ m/s	$L_{0l} = 107 + 27 \log[4(1 - \epsilon^2)\omega\rho_0 v(1 + l/2a)^2]$ dB		$L_{0h} = L_{0l} + 30$ dB
	$Z'_m = R_0$ $Z''_m = X_0(\delta)$ $R_0 = \frac{\rho_0}{\epsilon} \sqrt{8v\omega} \left(1 + \frac{l}{2a}\right) + (\rho_0/8\epsilon c_0)(2a\omega)^2$ $X_0 = \frac{\omega\rho_0}{\epsilon} \left[\sqrt{\frac{8v}{\omega}} \left(1 + \frac{l}{2a}\right) + l + \delta \right]$ $\delta = \delta_0 = 0.85(2a)\phi_0(\epsilon)$ $\phi_0(\epsilon) = 1 - 1.47\sqrt{\epsilon} + 0.47\sqrt{\epsilon^3}$	$Z'_m = \sqrt{R_h^2 - R_0^2}$ $Z''_m = X_0(\delta)$ $\delta = \delta_0\phi_1(M_0)$ $M_0 = \frac{10^{(-2.25+0.025L_p)}}{\sqrt{0.5\rho_0 c_0^2(1 + \epsilon^2)}}$	$Z'_m = R_h$ $Z''_m = X_0(\delta)$ $R_h = \frac{1}{\epsilon} \sqrt{2\rho_0(1 - \epsilon^2)} \times 10^{(-2.25+0.018L_p)}$ $\delta = \delta_0\phi_1(M_0)$ $\phi_1(M_0) = \frac{1 + 5 \times 10^3 M_0^2}{1 + 10^4 M_0^2}$
With flow, $M_\infty \geq 0.025$, $U_\infty \geq 10$ m/s	$L_p \leq L_{ml}$ $L_{ml} = 175 + 40 \log M_\infty$ dB $Z'_m = R_m$ $Z''_m = X_0(\delta)$ $R_m = 0.6 \rho_0 c_0 \frac{1 - \epsilon^2}{\epsilon} (M_\infty - 0.025) - 40R_0(M_\infty - 0.05); M_\infty \leq 0.05$ $R_m = 0.3 \rho_0 c_0 \frac{1 - \epsilon^2}{\epsilon} M_\infty$ $\delta = \delta_0\phi_2(M_\infty)$ $\phi_2(M_\infty) = 1/(1 + 305M_\infty^3)$	$L_{ml} \leq L_p \leq L_{mh}$ $Z'_m = \sqrt{R_m^2 + R_h^2}$ $Z''_m = X_0(\delta)$ $M_\infty > 0.05$ $\delta = \delta_0\phi_2(M_\infty)$	$L_p > L_{mh}$ $L_{mh} = L_{ml} + 18$ dB $Z'_m = R_h$ $Z''_m = X_0(\delta)$ $\delta = \delta_0\phi_1(M_\infty)$

^a L_p is octave-band SPL in dB re 20 μ Pa at the peak of the spectrum.

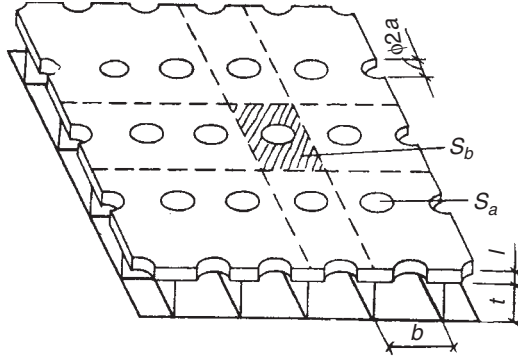


FIGURE 8.22 Resonators arranged on a raster to form a surface array.

Placing of flow-resistive materials in or behind the resonator opening reduces nonlinearity and sensitivity to grazing flow.

Spatial Average Impedance of Resonator Arrays

Locally reacting absorbers such as resonators are frequently used in the form of a surface array, shown in Fig. 8.22, where the individual resonators are arranged periodically. If the dimension b or $\sqrt{S_b}$ in Fig. 8.22 is much smaller than the acoustical wavelength, it is not necessary to take into account the interaction of the individual elements (resonators) with the sound field. To characterize the absorber, it is sufficient in this case to compute the spatial average wall impedance Z_1 :

$$Z_1 = \frac{\langle p \rangle}{\langle v \rangle} = \frac{S_b}{S_a} \frac{p_a}{v_a} = \frac{S_b}{S_a} Z_R = \frac{Z_R}{\varepsilon} \quad \text{N} \cdot \text{s}/\text{m}^3 \quad (8.65)$$

where Z_R is the impedance of the resonator as given in Eqs. (8.55) “measured” in the orifice of the resonator and $\varepsilon = S_a/S_b$ is the surface porosity. The effective wall impedance Z_1 is then used in Eq. (8.37) to determine the absorption coefficient. The impedances Z_m in Table 8.8 are effective impedances averaged over the plate surface (resonator mass impedance divided by porosity).

8.8 PLATE AND FOIL ABSORBERS

Plate absorbers are used in absorbing low-frequency tonal noise in situations where Helmholtz resonators are not feasible. The resonator consists of the mass of a thin plate and usually the stiffness of the air space between the plate and the elastic mounting at the perimeter of the plate. Foils are a special case of a thin limp plate.

Limp Thin Plate or Foil Absorber

The resonance frequency of a foil absorber is

$$f_0 = \frac{\sqrt{\rho_0 c_0^2 / m'' t}}{2\pi} \quad (8.66)$$

where t is the depth of the partitioned air space in meters and m'' is the mass per unit area of the limp foil in kilograms per square meter. If the air space depth is not small ($t > \frac{1}{8}\lambda$), there will be many resonance frequencies ω_n ($n = 0, 1, 2, \dots$) given by

$$\frac{\omega_n t}{c_0} \tan \frac{\omega_n t}{c_0} = \frac{\rho_0 t}{m''} \quad (8.67)$$

For an unpartitioned air space behind the foil the resonance frequency depends on the angle of sound incidence θ_0 . In this case t must be replaced in Eqs. (8.66) and (8.67) by $t \cos \theta_0$. If the air space is filled (without obstructing the movement of the foil) with a porous sound-absorbing material, then $(t/c_0)\Gamma_a''$ is used instead of t/c_0 in Eq. (8.67).

Foil-Wrapped Porous Absorber

Sound absorbers consisting of a protective layer of thin foil, a porous layer, and an air space behind are frequently used where the porous material must be protected from dust, dirt, and water. The effect of the thin foil can be taken into account by a series impedance $Z_s = j\omega m''$ that is added to the surface impedance Z_1 [see Eq. (8.37)]. It is essential that the foil is not stretched so that its inertia and not its membrane stress controls its response. Inserting a large-mesh (≥ 1 cm) wire cloth between the porous sound-absorbing material and the protective foil is a practical way to assure this.

Elastically Supported Stiff Plate

A form of resonant plate absorber that can be easily treated analytically is the elastically supported stiff plate. The elastic support may be localized by discrete points to occur along the perimeter in the form of a resilient gasket strip. The effective stiffness of such a resonator is

$$s'' = \frac{s_e}{S_p} + \frac{\rho_0 c_0^2}{t} \quad \text{N/m}^3$$

and the resonance frequency is

$$f_0 = \frac{1}{2\pi} \sqrt{\frac{s''}{m''}} = \frac{1}{2\pi} \sqrt{\frac{s_e/S_p + \rho_0 c_0^2/t}{m''}} \quad \text{Hz} \quad (8.68)$$

where $s_e = S_p \Delta p / \Delta x$ is the dynamic stiffness of resilient plate mounting and S_p is the surface area of the plate. Note that the elastic mounting increases the

resonance frequency (compared with that obtainable with a limp foil of the same m''). This can be compensated for by an appropriate increase of m'' . However, an increase of m'' increases the Q of the resonator ($Q = \sqrt{s''/m''}/R$) resulting in a narrower bandwidth.

Elastic Foil Absorber

Elastic foil absorbers consist of small air volumes enclosed in thin (200–400- μm) foil coffers. The coffers have typical dimensions of a few centimeters. A typical

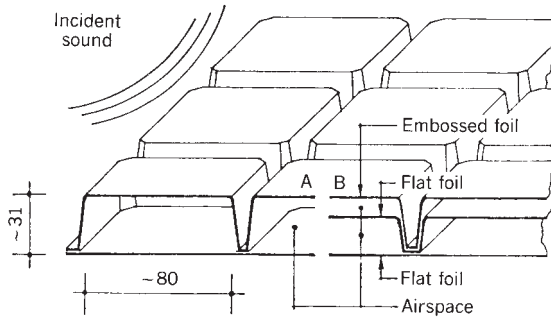


FIGURE 8.23 Construction of a foil absorber made of cold-drawn PVC foil.

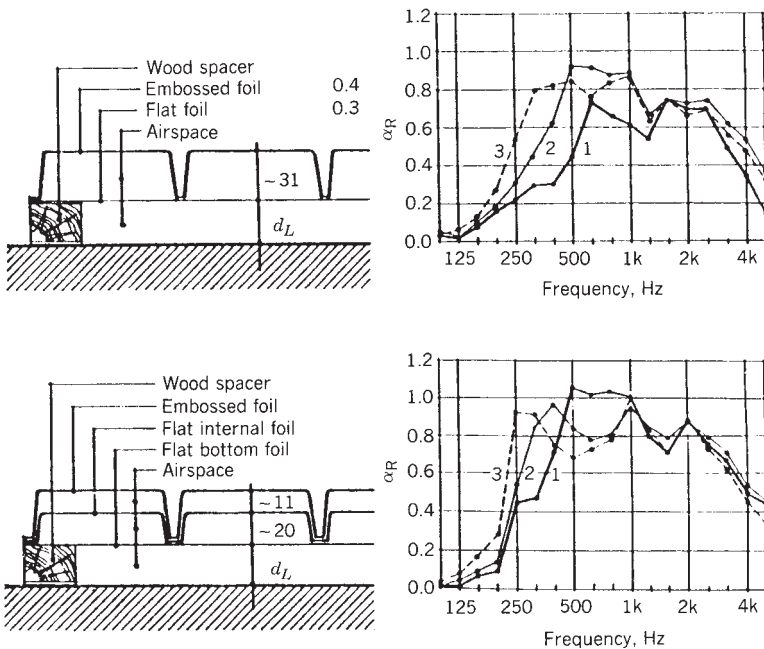


FIGURE 8.24 Random-incidence sound absorption coefficient α_R of absorbers made of cold-drawn PVC foil: (1) $d_L = 0$; (2) $d_L = 25$ mm; (3) $d_L = 50$ mm.

foil absorber is shown in Fig. 8.23. The wavelength of the incident sound is large compared with the typical dimensions of the coffers. Consequently, the incident sound periodically compresses the air in the individual coffers by exciting all of the volume-displacing modes of vibration of the various coffer walls. Because of the small thickness of the coffer walls, these resonances fall into the frequency range of 200–3150 Hz, which is of primary interest in building applications.⁴⁹ The foil material can be plastic (PVC) or metal. It is beneficial to emboss the foil because this leads to a more even distribution of the resonance frequencies. Figure 8.24 shows the measured random-incidence sound absorption coefficient obtained with two different foil absorber configurations, indicating that significant broadband sound absorption is achievable.

The key advantage of these foil absorbers is that they are nonporous, do not support bacterial growth, are lightweight, and can be made light transparent. Their main application is in breweries, packaging plants, hospitals, and computer chip manufacturing areas, where high emphasis is placed on hygiene and dust cannot be tolerated. They also lend themselves as sound absorbers in high-moisture environments such as swimming pools and as muffler baffles in cooling towers. Their light transparency is a distinct advantage in industrial halls. Considerable experience has been gained in such applications in Europe.⁴⁹

REFERENCES

1. K. U. Ingard, *Notes on Sound Absorption Technology*, Version 94-02, Noise Control Foundation, Poughkeepsie, NY, October 1994.
2. F. P. Mechel, *Schall Absorber [Sound Absorbers]*, Vol. 1: *Aeusserer Schallfelder, Wechselwirkungen [Outer Sound Fields, Interactions]*; Vol. 2: *Innere Schallfelder, Strukturen [Internal Sound Fields, Structures]*; Vol. 3: *Anwendungen [Application]*, S. Hirzel Verlag, Stuttgart, 1989.
3. F. P. Mechel, *MAPS-Mechels Acoustic Program System*, CD-ROM for Windows, S. Hirzel, Stuttgart, Germany, 2001.
4. H. Kuttruff, *Room Acoustics*, Applied Science, London, 1973.
5. L. L. Beranek, *Acoustic Measurements*, Acoustical Society of America, Melville, New York, 1988.
6. R. D. Godfrey, Owens Corning Fiberglass Company, private communication, June 2004.
7. T. Northwood, *J. Acoust. Soc. Am.*, **31**, 596 (1959); **35**, 1173 (1963).
8. M. J. Swift, "The Physical Properties of Porous Recycled Materials," Ph. D. Thesis, University of Bradford, United Kingdom, 2000.
9. L. Cremer and M. Hubert, *Vorlesungen Ueber Technische Akustik [Lectures on Technical Acoustics]*, 3rd ed., Springer, Berlin, 1985.
10. Y. Champoux, M. R. Stinson, and G. A. Daigle, "Air-Based System for the Measurement of Porosity," *J. Acoust. Soc. Am.*, **89**, 910–916 (1991).

11. P. Leclaire, O. Umnova, K. V. Horoshenkov, and L. Maillet, "Porosity Measurement by Comparison of Air Volumes," *Rev. Sci. Instrum.*, **74**, 1366–1370 (2003).
12. Z. E. A. Fellah, S. Berger, W. Lauriks, C. Depollier, C. AristBgui, and J.-Y. Chapelon, "Measuring the Porosity and the Tortuosity of Porous Materials via Reflected Waves at Oblique Incidence," *J. Acoust. Soc. Am.*, **113**, 2424–2433 (2003).
13. Z. E. A. Fellah, S. Berger, W. Lauriks, C. Depollier, and M. Fellah, "Measuring the Porosity of Porous Materials Having a Rigid Frame via Reflected Waves: A Time Domain Analysis with Fractional Derivatives," *J. Appl. Phys.*, **93**, 296–303 (2003).
14. Z. E. A. Fellah, S. Berger, W. Lauriks, C. Depollier, and J. Y. Chapelon, "Inverse Problem in Air-Saturated Porous Media via Reflected Waves," *Rev. Sci. Int.*, **74**, 2871–2879 (2003).
15. Z. E. A. Fellah, S. Berger, W. Lauriks, C. Depollier, W. Lauriks, P. Trompette, and J. Y. Chapelon, "Ultrasonic Measurement of the Porosity and Tortuosity of Air-Saturated Random Packings of Beads," *J. Appl. Phys.* to appear.
16. A. S. Sangani and A. Acrivos, "Slow Flow through Periodic Array of Spheres," *Int. J. Multiphase Flow*, **8**, 343–360 (1982).
17. W. S. Jodrey and E. M. Tory, "Computer-Simulation of Close Random Packing of Equal Spheres," *Phys. Rev. A*, **32**, 2347–2351 (1985).
18. N. Sen, C. Scala, and M. H. Cohen, "A Self-Similar Model for Sedimentary Rocks with Application to the Dielectric Constant of Fused Glass Beads," *Geophysics*, **46**, 781–795 (1981).
19. K. Attenborough, "Acoustical Characteristics of Rigid Absorbents and Granular Media," *J. Acoust. Soc. Am.*, **83**(3), 785–799 (March 1983).
20. O. Umnova, K. Attenborough, and K. M. Li, "Cell Model Calculations of Dynamic Drag Parameters in Packings of Spheres," *J. Acoust. Soc. Am.*, **107**(3), 3113–3119 (2000).
21. J. F. Allard, *Propagation of Sound in Porous Media*, Elsevier Applied Science, London, 1993.
22. J. F. Allard, B. Castagnede, M. Henry, and W. Lauriks, "Evaluation of Tortuosity in Acoustic Porous Materials Saturated by Air," *Rev. Sci. Instrum.*, **65**, 754–755 (1994).
23. ASTM C 522-87, "Standard Test Method for Airflow Resistance of Acoustical Materials," American Society for Testing and Materials, West Conshohocken, PA, pp. 169–173.
24. M. R. Stinson and G. A. Daigle, "Electronic System for the Measurement of Flow Resistance," *J. Acoust. Soc. Am.*, **83**, 2422–2428 (1988).
25. O. Umnova, University of Salford, United Kingdom, September, 2004. private communication.
26. P. Leclaire, M. J. Swift, and K. V. Horoshenkov. "Determining the Specific Area of Porous Acoustic Materials from Water Extraction Data," *J. Appl. Phys.*, **84**, 6886–6890 (1998).
27. K. Attenborough, "Models for the Acoustical Properties of Air-Saturated Granular Media," *Acta Acustica*, **1**, 213–226 (1993).
28. K. V. Horoshenkov, K. Attenborough, and S. N. Chandler-Wilde, "Pade Approximants for the Acoustical Properties of Rigid Frame Porous Media with Pore Size Distribution," *J. Acoust. Soc. Am.*, **104**, 1198–1209 (1998).

29. K. V. Horoshenkov and M. J. Swift, "The Acoustic Properties of Granular Materials with Pore Size Distribution Close to Log-normal," *J. Acoust. Soc. Am.*, **110**, 2371–2378 (2001).
30. D. L. Johnson, J. Koplik, and R. Dashen, "Theory of Dynamic Permeability and Tortuosity in Fluid-Saturated Porous Media," *J. Fluid Mech.*, **176**, 379–402 (1987).
31. O. Umnova, K. Attenborough, and K. M. Li, "A Cell Model for the Acoustical Properties of Packings of Spheres," *Acustica combined with Acta Acustica*, **87**, 26–235 (2001).
32. J. F. Allard and Y. Champoux, "New Empirical Relations for Sound Propagation in Rigid Frame Fibrous Materials," *J. Acoust. Soc. Am.*, **91**, 3346–3353 (1992).
33. N. N. Voronina and K. V. Horoshenkov, "A New Empirical Model for the Acoustic Properties of Loose Granular Media," *Appl. Acoust.*, **64**(4), 415–432 (2003).
34. J. Y. Chung and D. A. Blaser, "Transfer Function Method of Measuring In-Duct Acoustic Properties. I. Theory, II Experiment," *J. Acoust. Soc. Am.*, **68**(3), 907–921 (1980).
35. C. D. Smith, and T. L. Parrott, "Comparison of Three Methods for Measuring Acoustic Properties of Bulk Materials," *J. Acoust. Soc. Am.*, **74**(5), 1577–1582 (1983).
36. H. Utsuno et al., "Transfer Function Method for Measuring Characteristic Impedance and Propagation Constant of Porous Materials," *J. Acoust. Soc. Am.*, **86**(2), 637–643 (1989).
37. P. Mechel, *Akustische Kennwerte von Faserabsorbem [Acoustic Parameters of Fibrous Sound Absorbing Materials]*, Vol. I, Bericht BS 85/83; Vol. II, Bericht BS 75/82, Fraunhofer Inst. Bauphysik, Stuttgart, Germany.
38. M. E. Delany and E. N. Bazley, "Acoustical Properties of Fibrous Absorbent Materials," *Appl. Acoust.*, **3**, 105–116 (1970).
39. Y. Miki, "Acoustical Properties of Porous Materials—Modifications of Delany-Bazley Models," *J. Acoust. Soc. Jpn. (E)*, **11**, 19–24 (1990).
40. M. Garai and F. Pompoli, "A Simple Empirical Model of Polyester Fibre Materials for Acoustical Applications," *Appl. Acoust.* (in press).
41. A. Cummings and S. P. Beadle, "Acoustic Properties of Reticulated Plastic Foams," *J. Sound Vib.*, **175**(1), 115–133 (1993).
42. Q. Wu, "Empirical Relations between Acoustical Properties and Flow Resistivity of Porous Plastic Open-Cell Elastic Foams," *Appl. Acoust.*, **25**, 141–148 (1988).
43. L. P. Dunn and W. A. Davern "Calculation of Acoustic Impedance of Multilayer Absorbers," *Appl. Acoust.*, **19**, 321–334 (1986).
44. R. J. Astley and A. Cummings, "A Finite Element Scheme for Attenuation in Ducts Lined with Porous Material: Comparison with Experiment," *J. Sound Vib.*, **116**, 239–263 (1987).
45. J. Royar, "Untersuchungen zum Akustischen Absorber-Kanten-Effekt an einem zwei-dimensionalen Modell" ["Investigation of the Acoustic Edge-Effect Utilizing a Two-Dimensional Model"], Ph.D. Thesis, Faculty of Mathematics and Nature Sciences, University of Saarbrücken, Germany, 1974.
46. F. P. Mechel, "Design Charts for Sound Absorber Layers," *J. Acoust. Soc. Am.*, **83**(3), 1002–1013 (1988).

47. F. Mechel, "Sound Absorbers and Absorber Functions," in G. L. Osipova and E. J. Judina, (Eds.), *Reduction of Noise in Buildings and Inhabited Regions* (in Russian), Strojnizdat, Moscow, 1987.
48. J. L. B. Coelho, "Acoustic Characteristics of Perforate Liners in Expansion Chambers," Ph.D. Thesis, Institute of Sound Vibration, Southampton, 1983.
49. F. Mechel and N. Kieseewetter, "Schallabsorberaus Kunststoff-Folie" ["Plastic-Foil Sound Absorber"], *Acustica*, **47**, 83–88 (1981).